

# BDA Course 2026

## Lecture 3

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# Outline Chapter 3

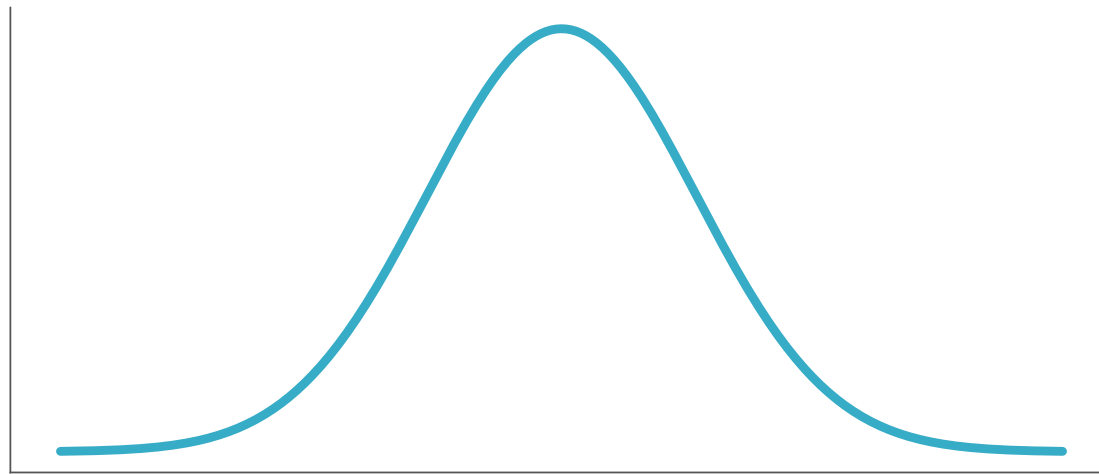
- 3.1 Marginalization
- 3.2 Normal distribution with a noninformative prior (important)
- 3.3 Normal distribution with a conjugate prior (important)
- 3.4 Multinomial model (can be skipped)
- 3.5 Multivariate normal with known variance (useful for chapter 4)
- 3.6 Multivariate normal with unknown variance (glance through)
- 3.7 Bioassay example (very important, related to one of the exercises)
- 3.8 Summary

# This lecture

- Normal / Gaussian distribution with unknown mean and scale
- Marginalization by analytic integration
- Posterior predictive distribution by analytic integration
- Generalized Linear Models (GLMs)
- Gaussian process (extra-material not covered in detail)

# Normal / Gaussian

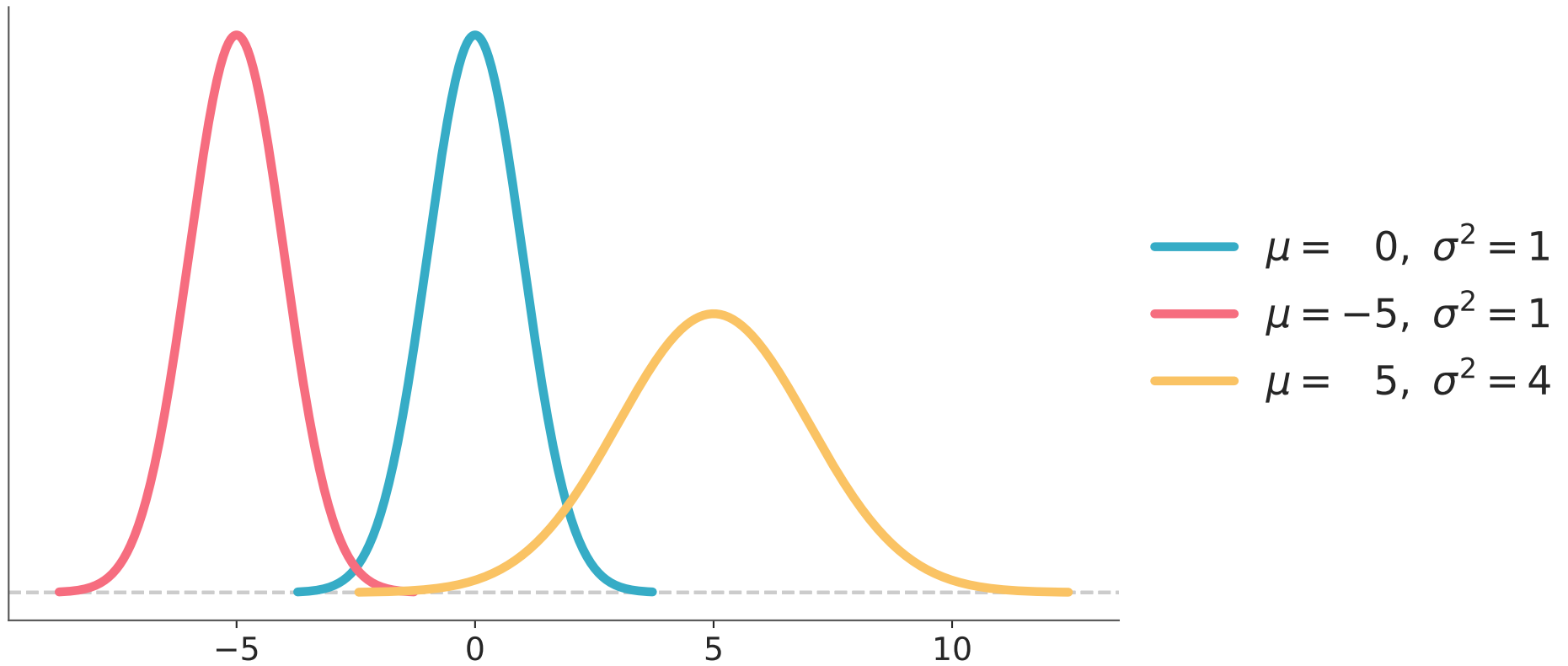
- Gaussian: favored by Germans (but it was studied earlier by De Moivre and Laplace, and Gauss avoided using it)
- Normal: some prominent statistician started to use this term systematically in the end of 19th century (but it's a bit of exaggeration)



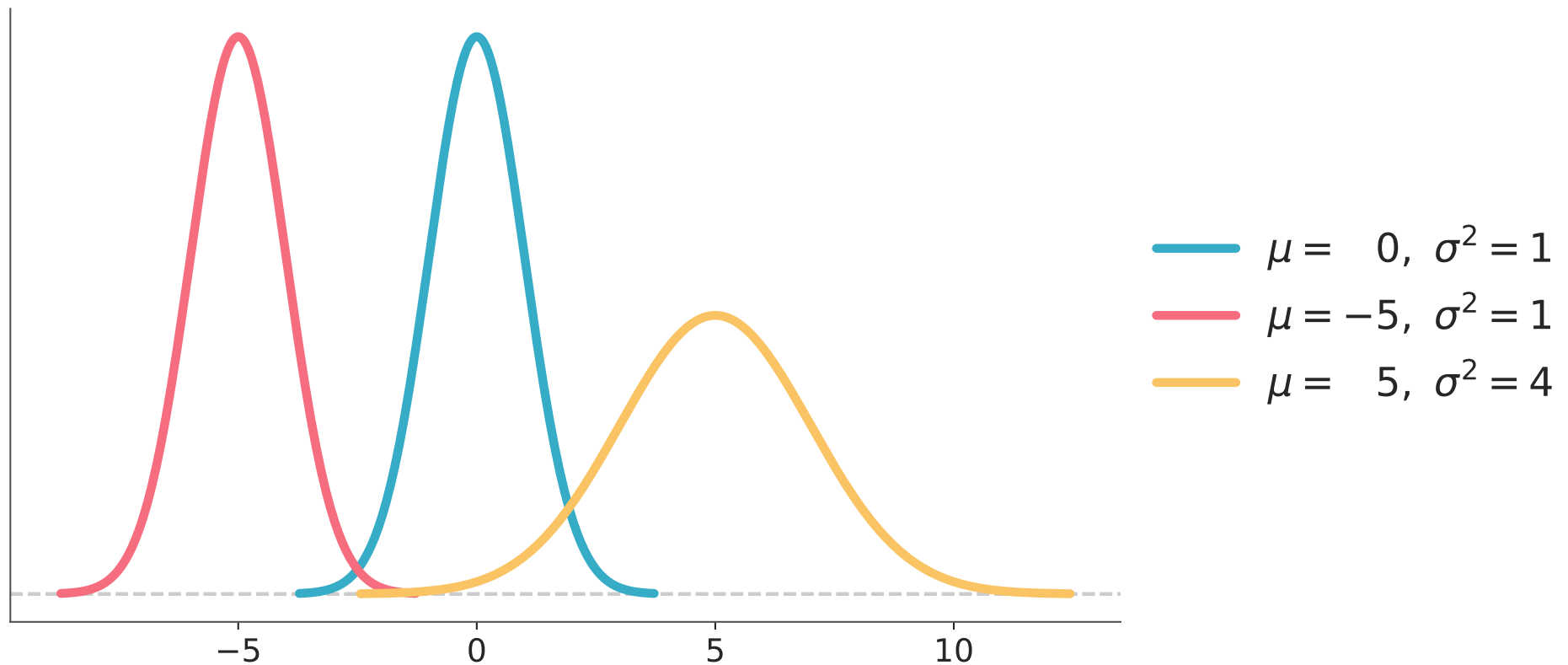
# Normal / Gaussian

- On its own
  - Analysis of real valued observations
  - Sometimes convenient approximation for discrete observations
  - Often convenient assumption
- As a building block of more complex models
  - Normal linear regression
  - Gaussian processes (non parametric, infinite-dimensional generalization of the normal distribution)
  - Kalman filters (recursive estimation for linear Gaussian state-space models)
- As an approximation tool for Bayesian inference
  - Posterior distribution approximation with Laplace, variational inference, expectation propagation

# Normal / Gaussian



# Normal / Gaussian

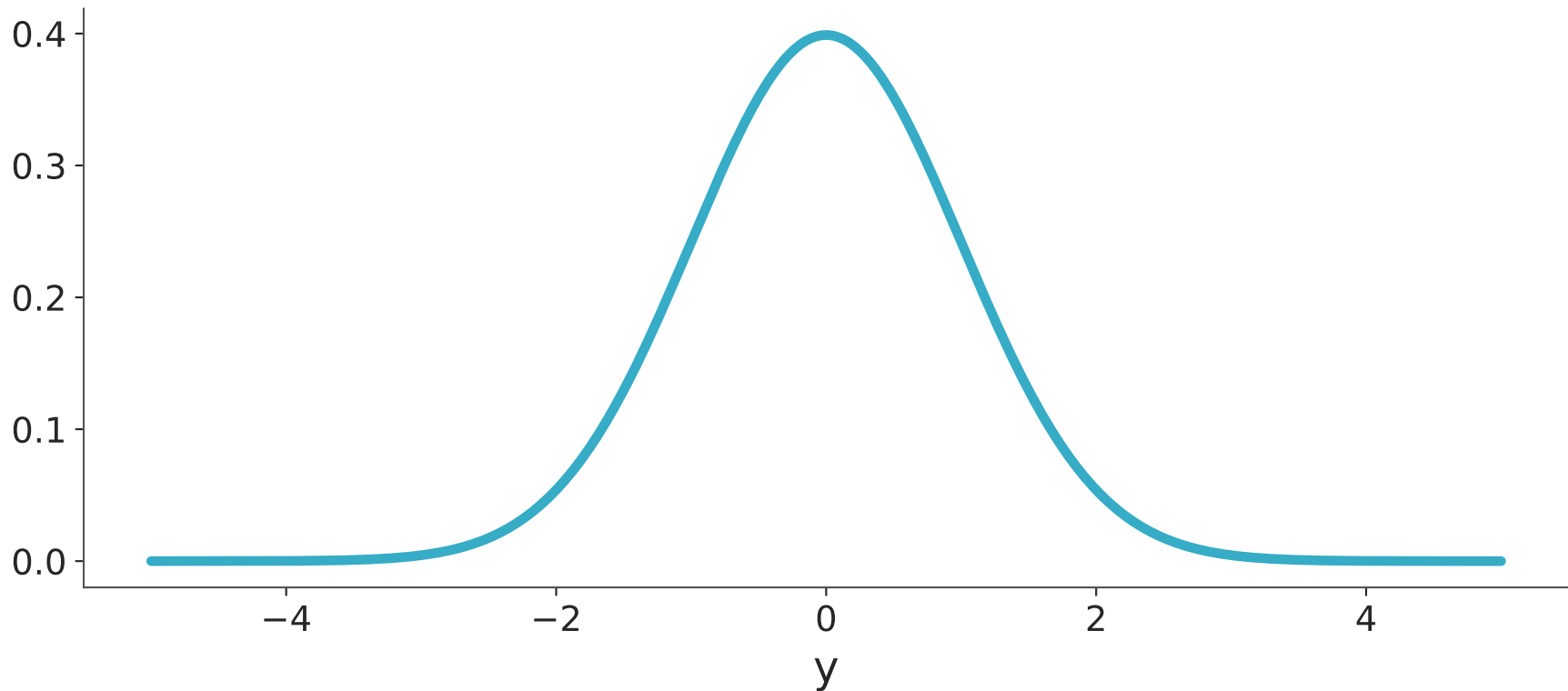


- $y \sim N(\mu, \sigma^2)$  with variance  $\sigma^2$  (useful in derivations)
- $y \sim \text{normal}(\mu, \sigma)$  with deviation  $\sigma$  (useful for interpretation, used in many probabilistic programming languages, like Stan)

# Normal / Gaussian

observation  $y$ , and parameters  $\mu$  and  $\sigma$

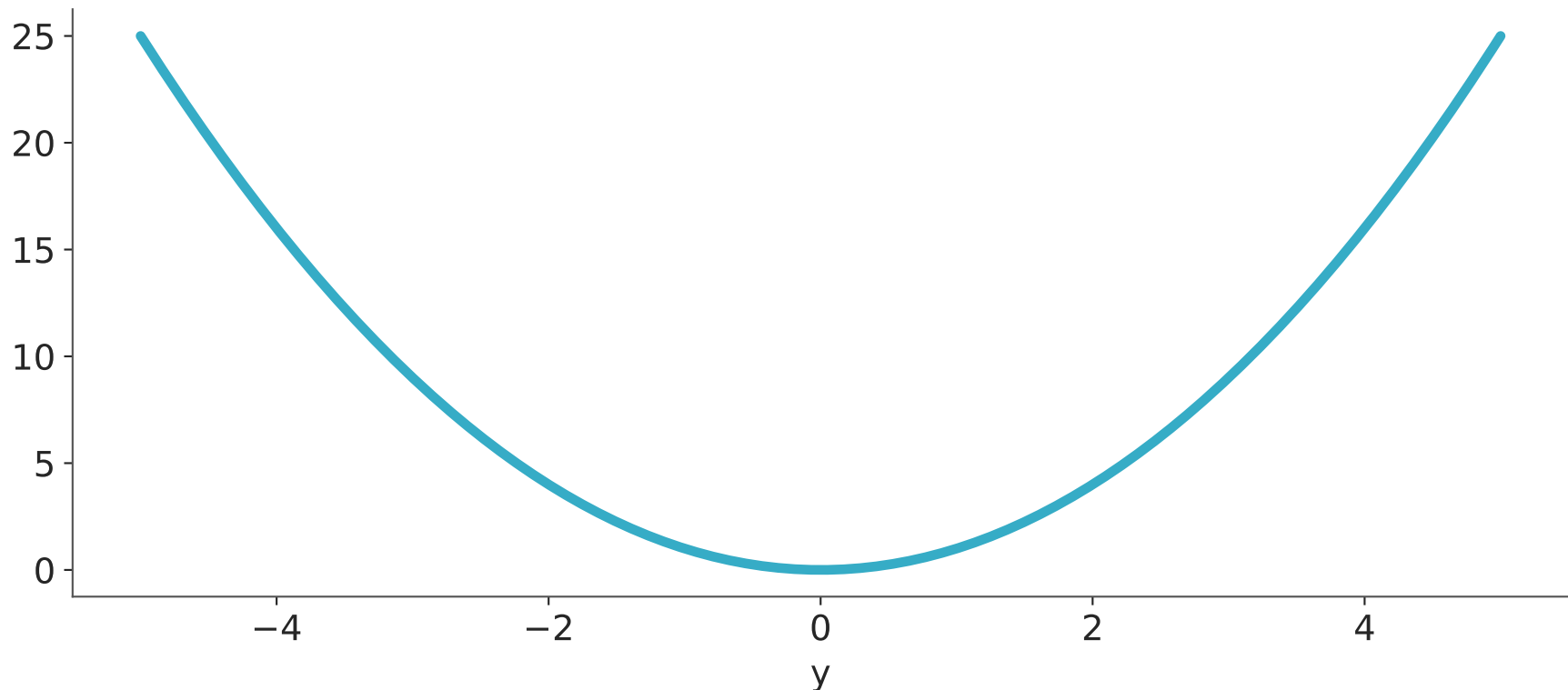
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



# Normal / Gaussian

- The square of the differences

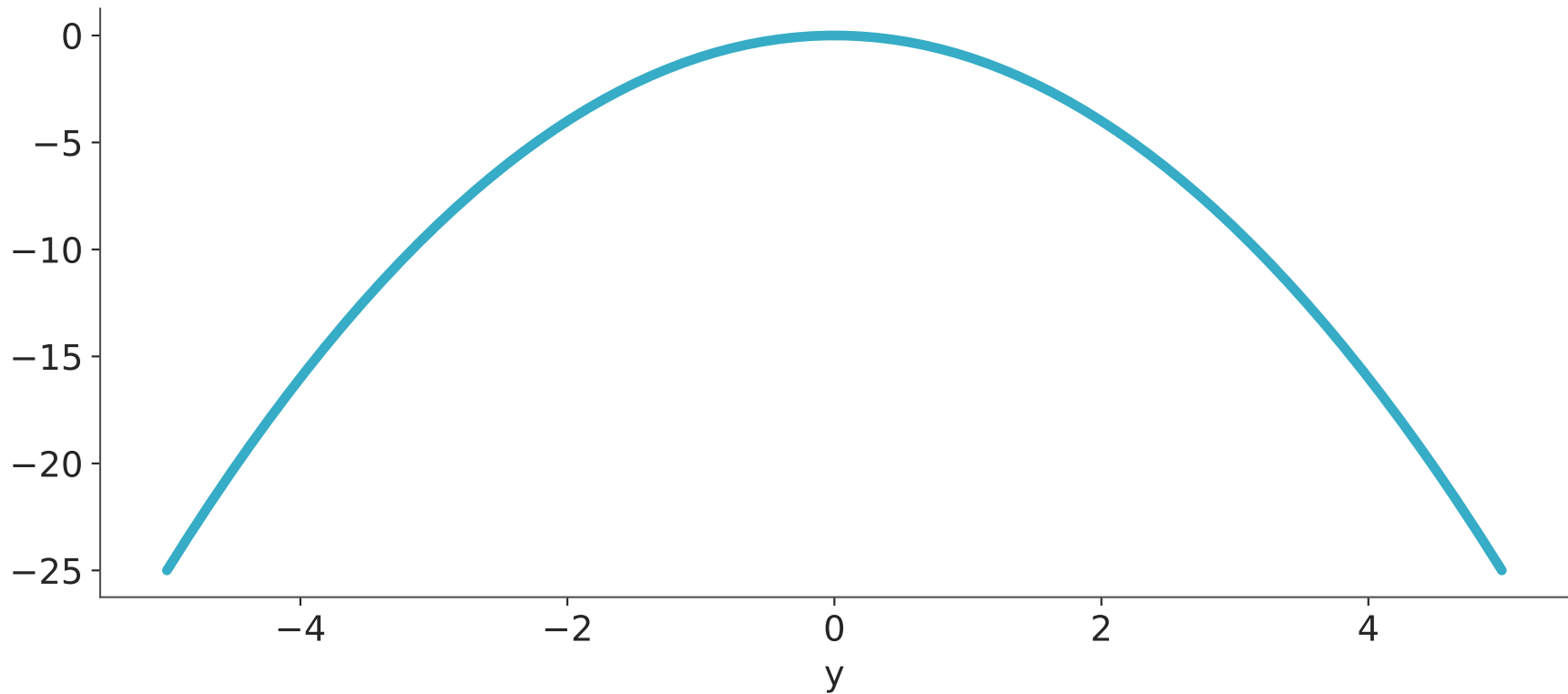
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



# Normal / Gaussian

- The minus square of the differences

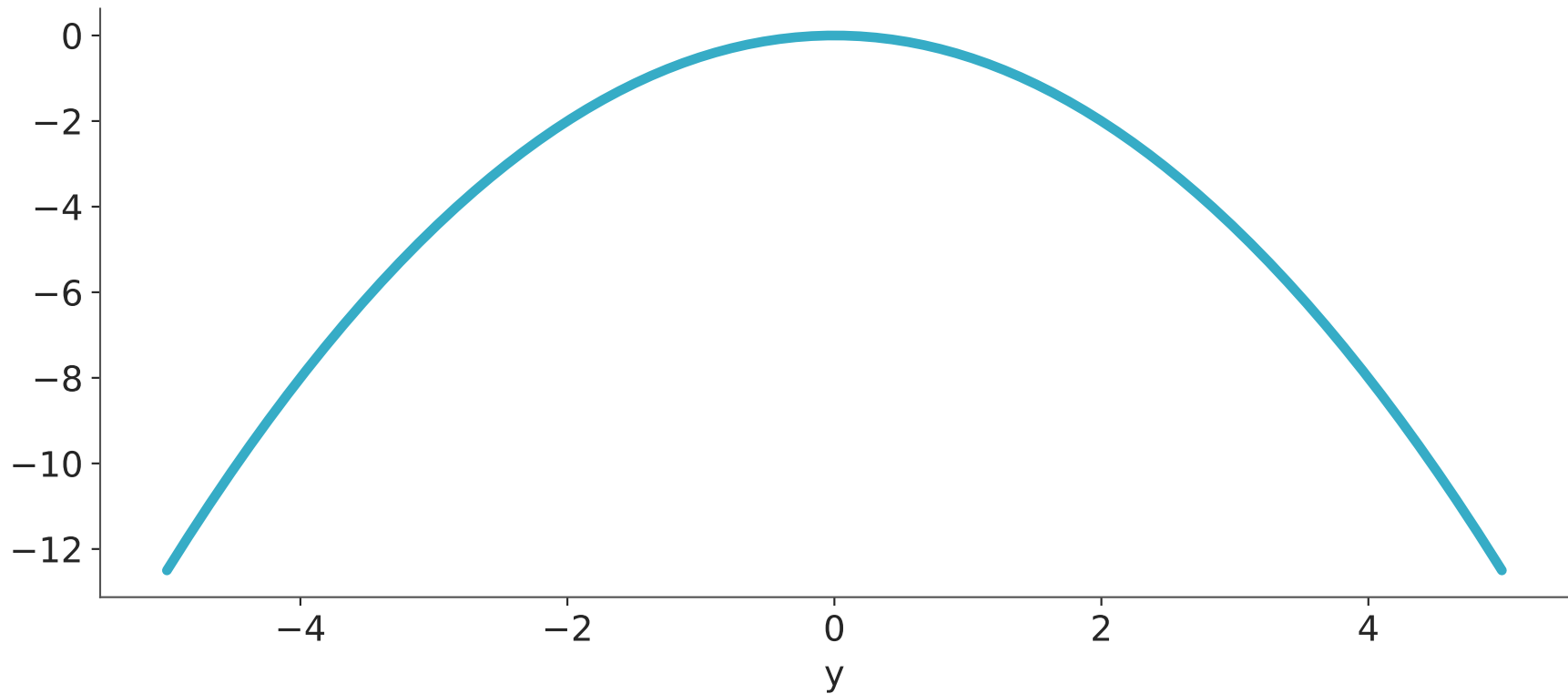
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# Normal / Gaussian

- The scaled minus square of the differences

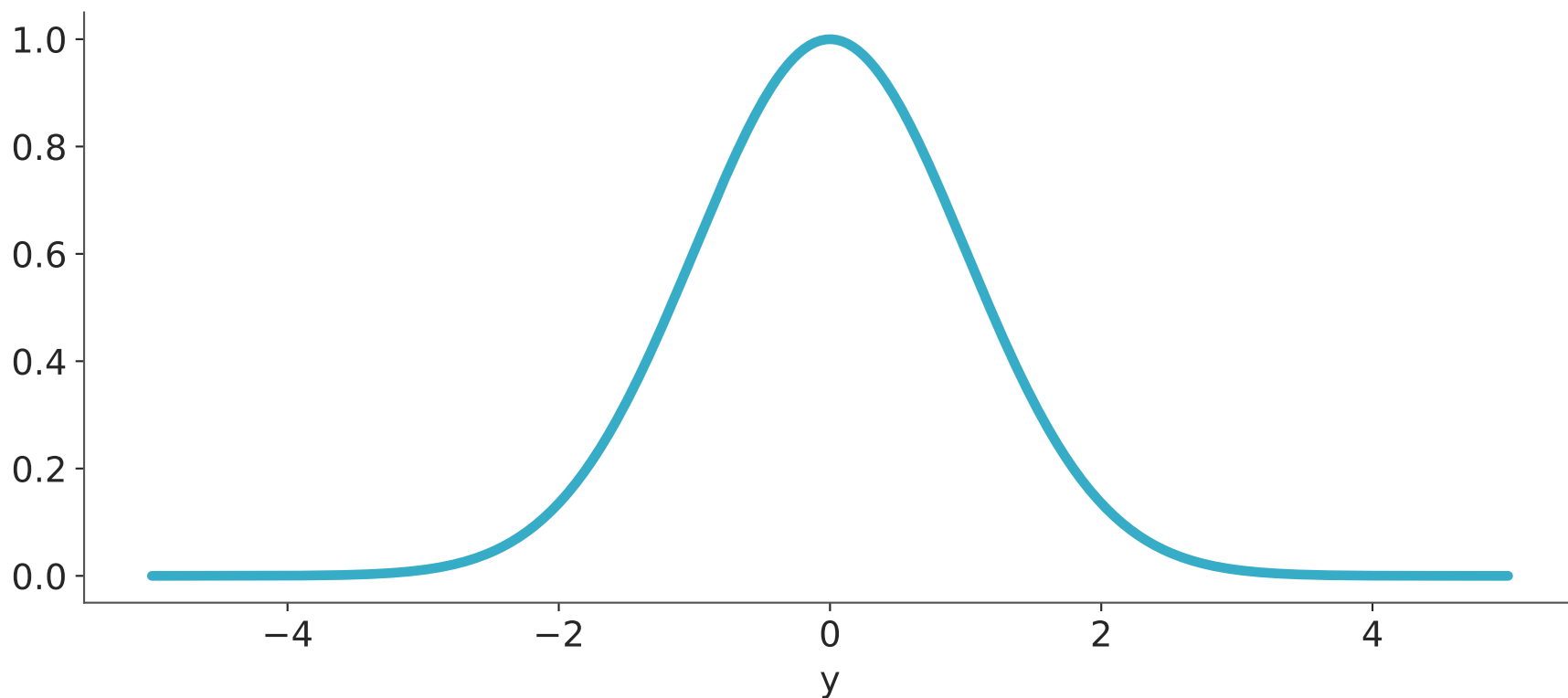
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



# Normal / Gaussian

- The exponentiated minus scaled square of the differences

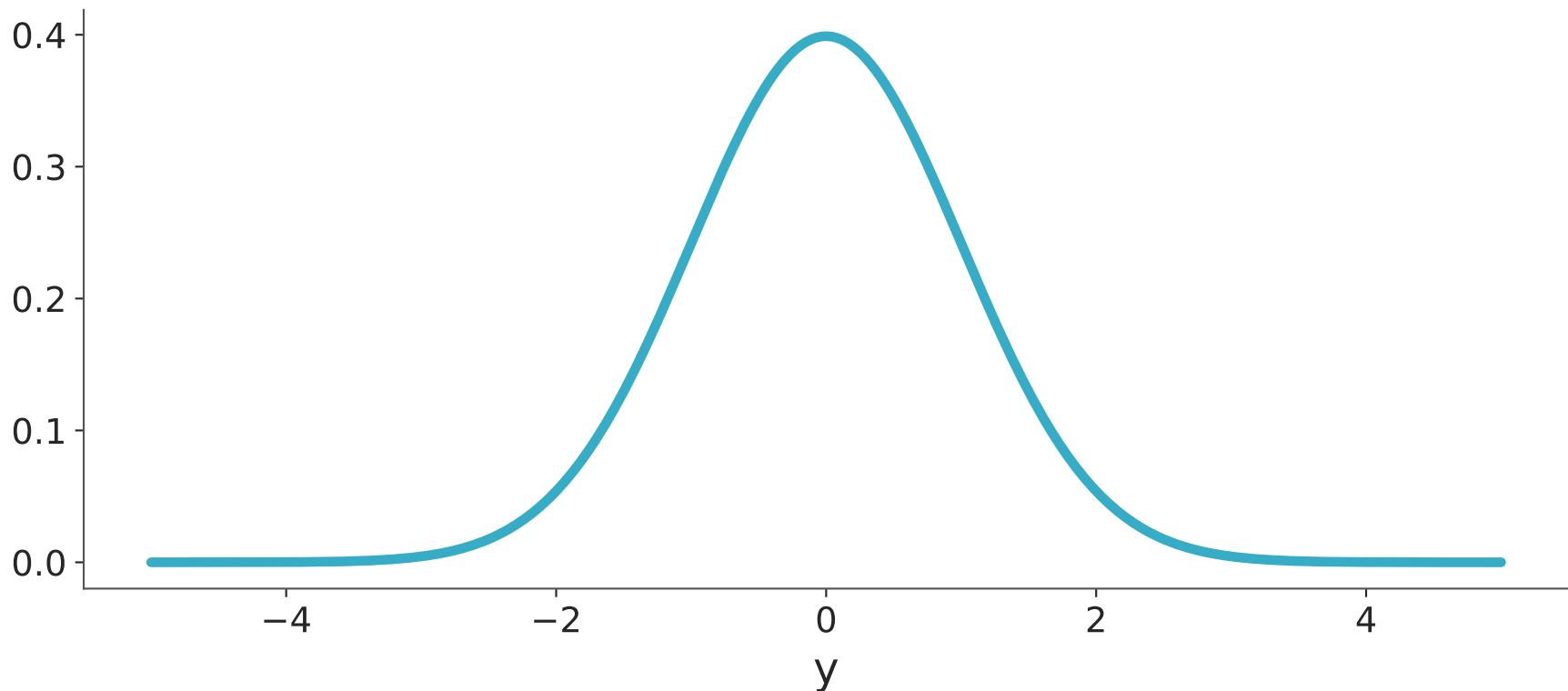
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



# Normal / Gaussian

- The normalized exponentiated minus scaled square of the differences

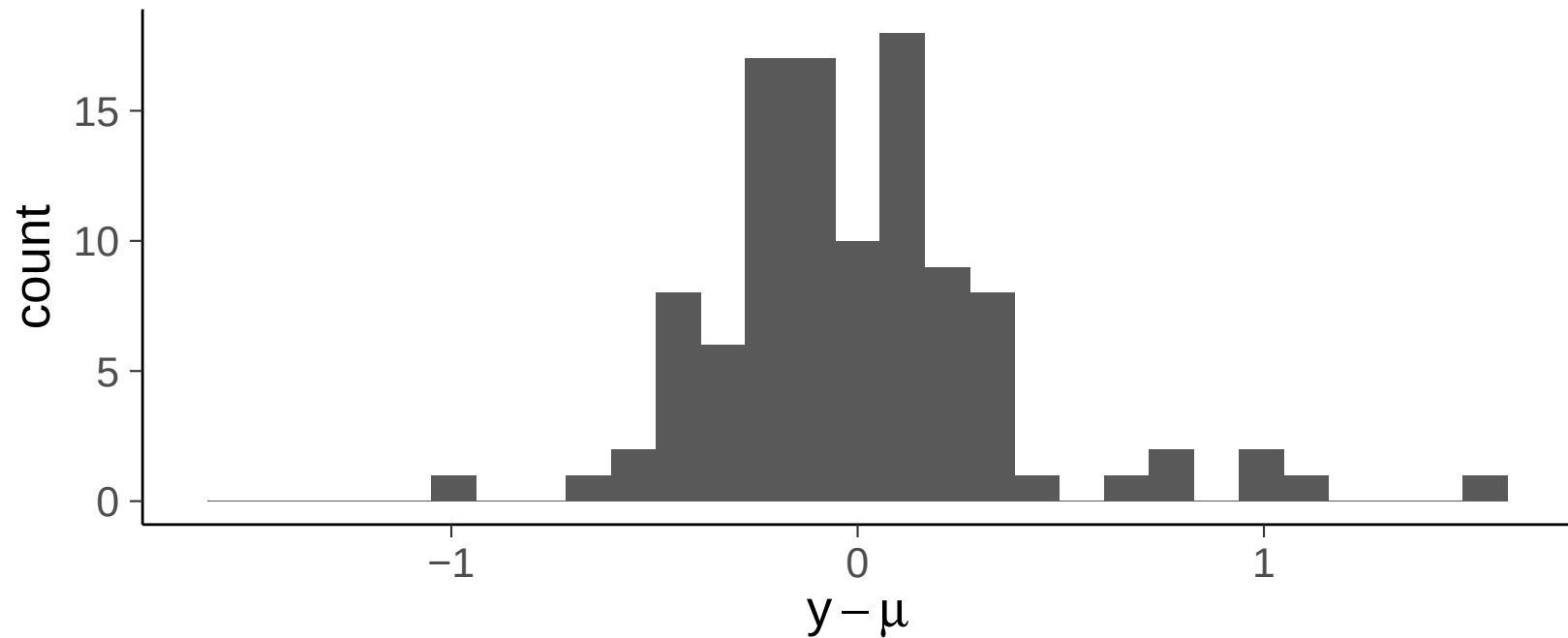
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Why  $\pi$  is here and how it is connected to CLT? → <https://www.youtube.com/watch?v=cy8r7WSuT1I>

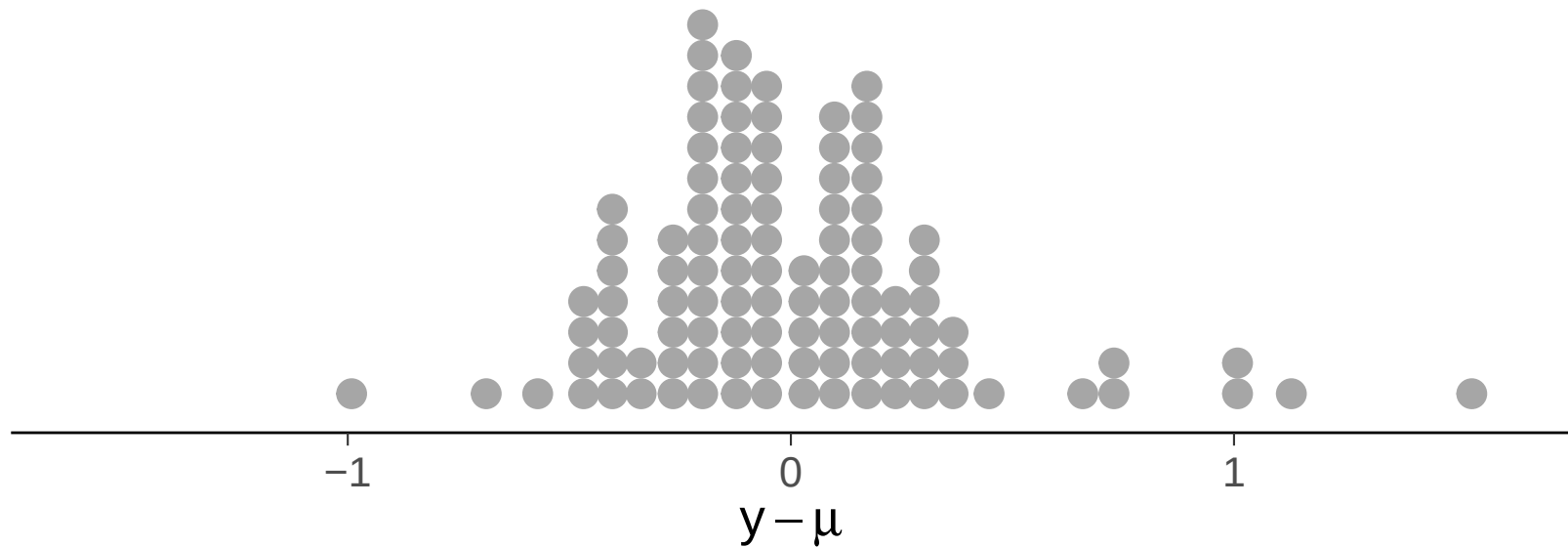
# Variation in helicopter flight times

- Take the mean of the helicopter flight times
- To each observation, subtract the mean to get the residuals
- Histogram of residuals



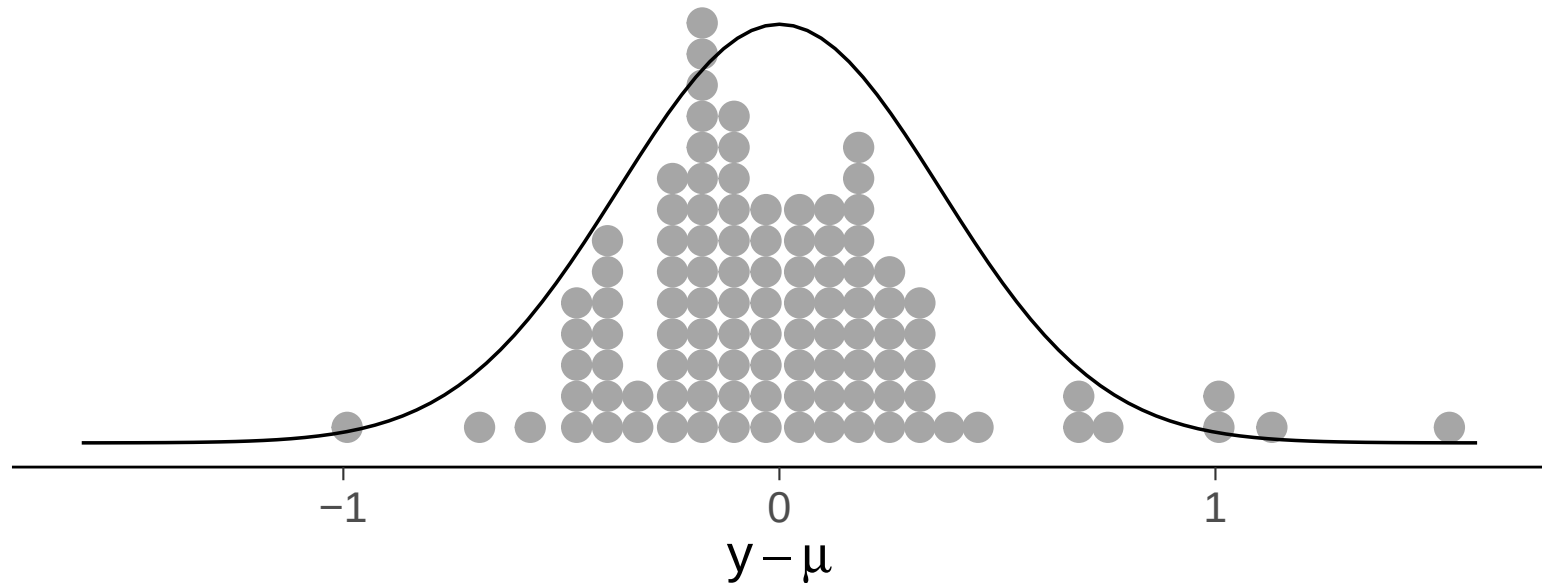
# Variation in helicopter flight times

- Quantile-dot plot.
  - Each dot represents an equal fraction of the observations.
  - In this example we have 100 dots, so each dot is 1% of the observations.



# Variation in helicopter flight times

- Quantile-dot plot + PDF of the normal distribution



- Why close to normally distributed?
  - Many small sources of variations summed together tend to be close to normally distributed
  - Central Limit Theorem (CLT)

# Normal data model and likelihood

- **Data model** for  $y$ :  $p(y \mid \mu, \sigma)$ , a distribution over  $y$  for fixed  $\mu, \sigma$

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- With  $n$  helicopter flight times  $y = (y_1, \dots, y_n)$  and assuming conditional independence.

$$p(y \mid \mu, \sigma) = \prod_{i=1}^n p(y_i \mid \mu, \sigma),$$

# Posterior for $\mu$ and $\sigma$ given $y$

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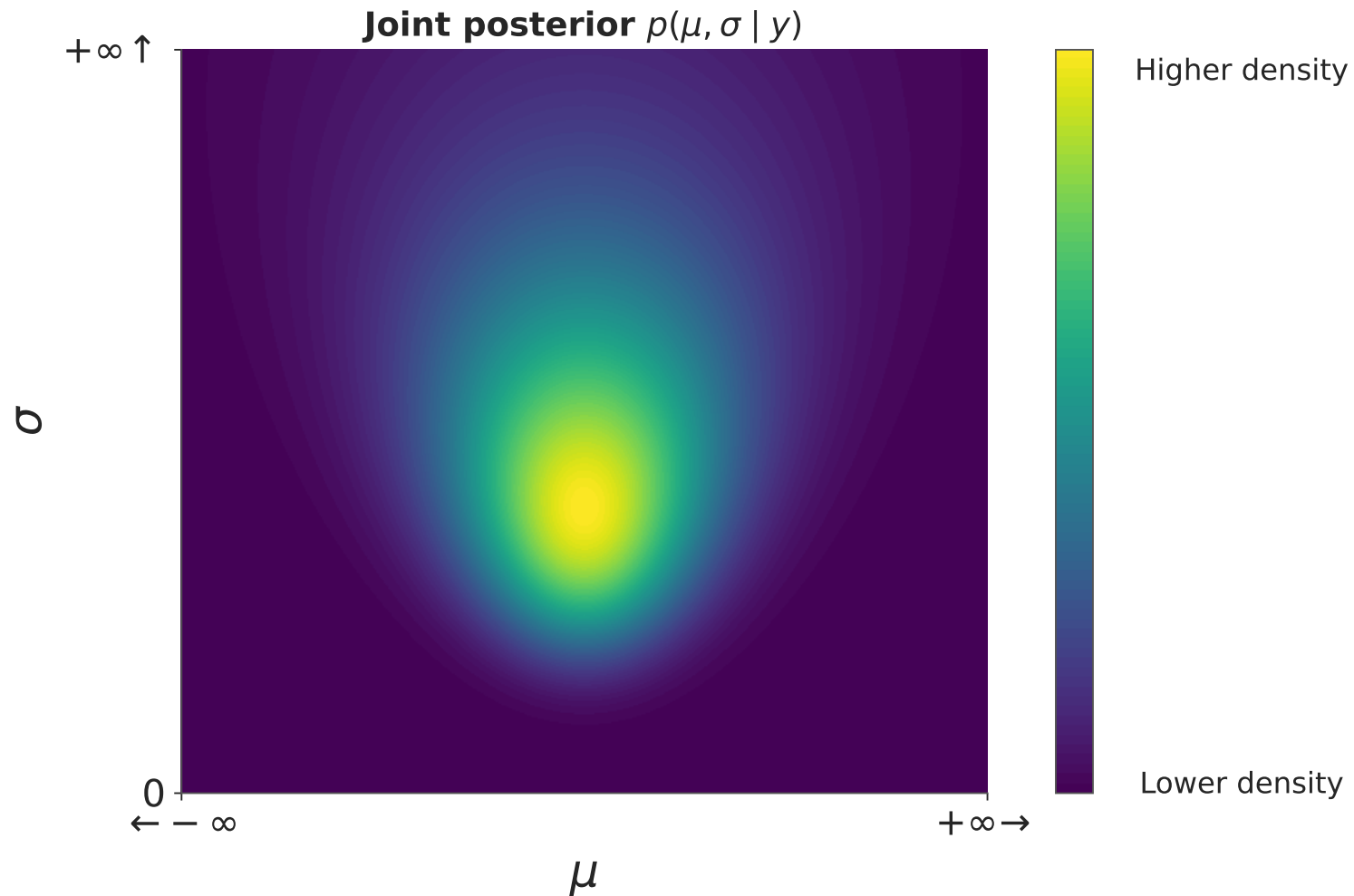
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- **Prior** for  $\mu$  and  $\sigma$ :
  - joint  $p(\mu, \sigma)$
  - independent  $p(\mu)p(\sigma)$
- Bayes' rule gives the joint **posterior**

$$p(\mu, \sigma \mid y) \propto \prod_{i=1}^n p(y_i \mid \mu, \sigma) \quad p(\mu, \sigma)$$

# Posterior for $\mu$ and $\sigma$ given $y$

- This posterior is 2-dimensional, covering both parameters jointly



- For some priors analytic solutions exist; for others we need numerical methods

# Non-informative prior for $\mu, \sigma$

- Assume independence:

$$p(\mu, \sigma) \propto p(\mu) \times p(\sigma)$$

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- We want  $p(\sigma)$ , not  $p(\log \sigma)$ . Change of variables (BDA3 p. 21):

$$p(\sigma) = p(\log \sigma) \left| \frac{d \log \sigma}{d\sigma} \right| = 1 \times \frac{1}{\sigma} = \sigma^{-1}$$

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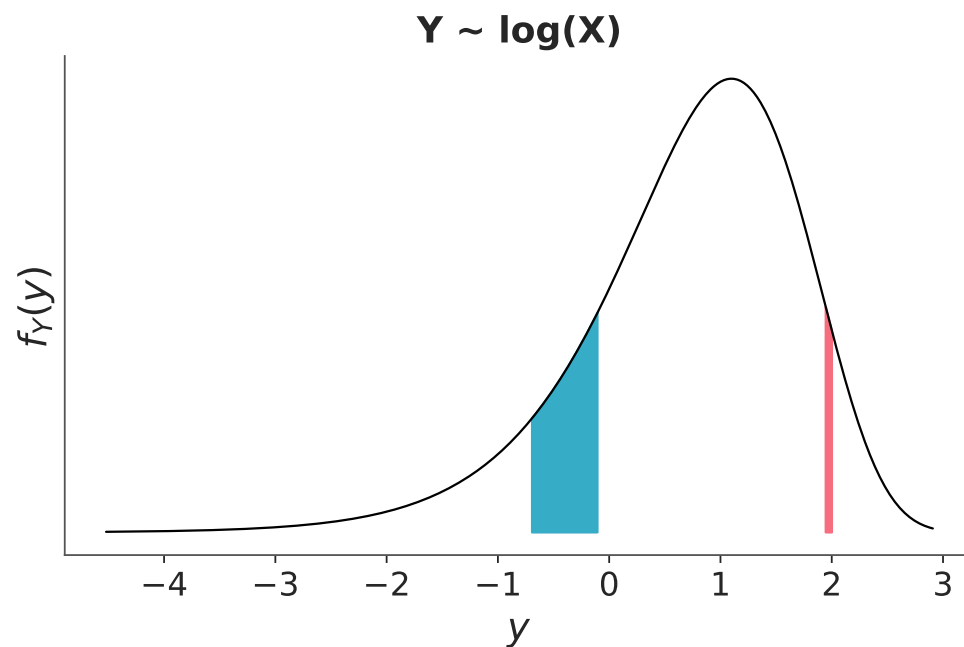
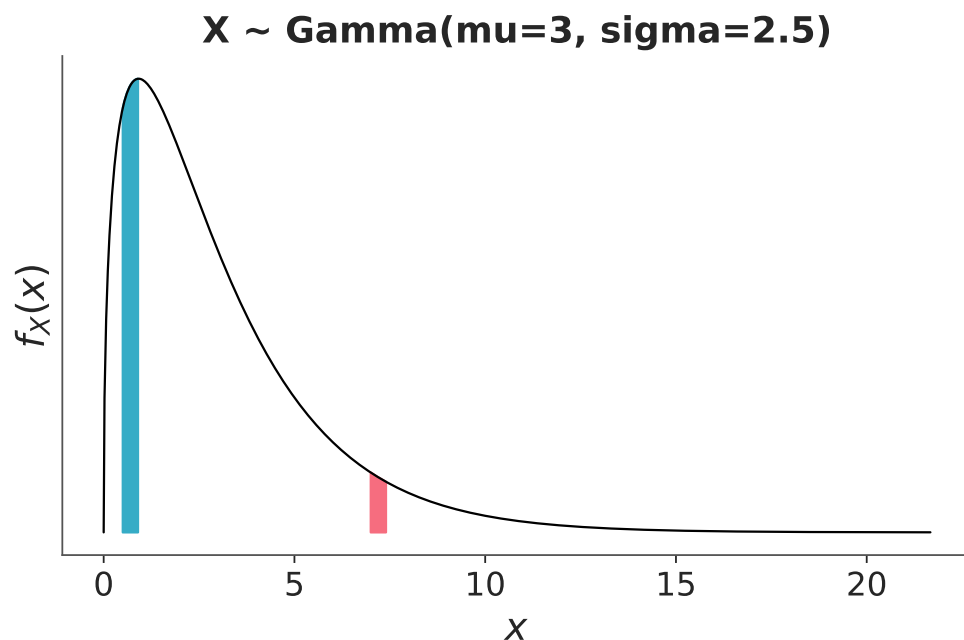
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- Putting it all together:

$$p(\mu, \sigma^2) \propto \sigma^{-2}$$

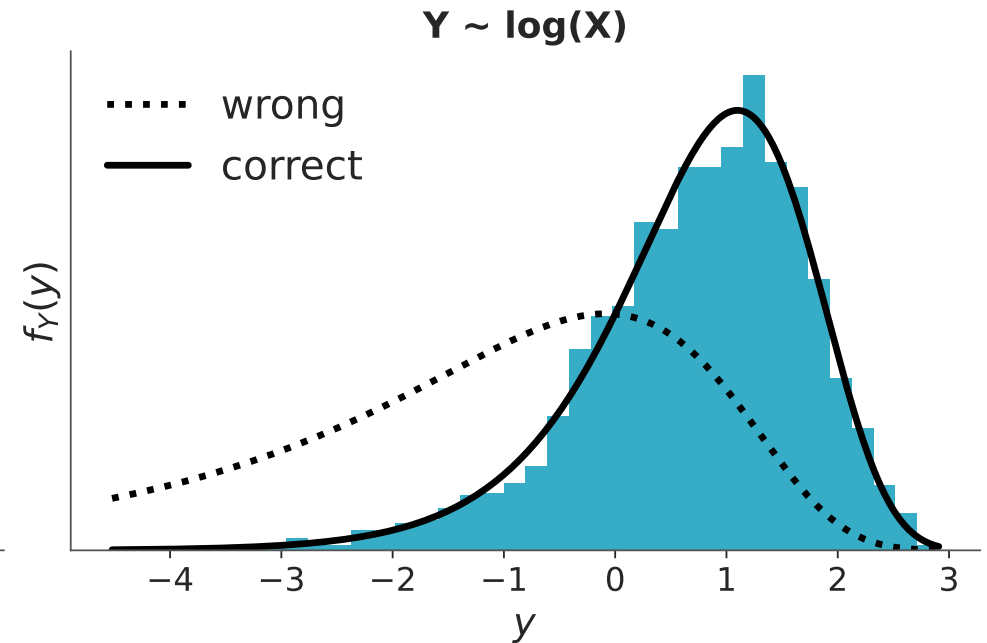
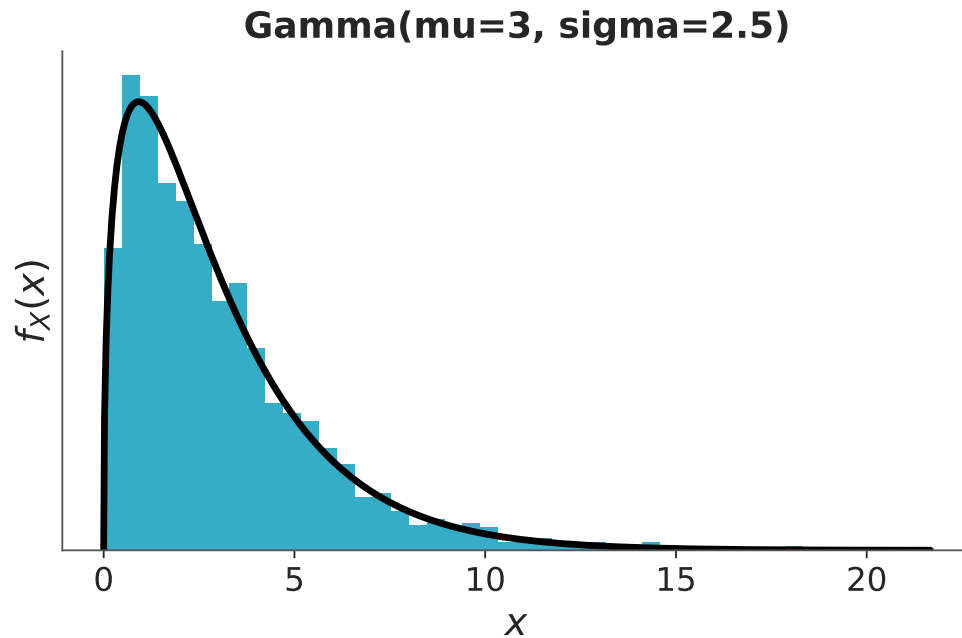
# Change of variable intuition

- Suppose  $X \sim \text{Gamma}(\alpha, \beta)$ , but we want a new variable  $Y = \log(X)$ .
- The transformation squishes and stretches intervals, so the density must adjust to preserve probability.



# Change of variable intuition

- A naive transformation  $p_{X(\exp(Y))}$  gives the wrong density for  $Y = \log(X)$ .



# Joint posterior

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- Substituting the non-informative prior and the normal density for each  $p(y_i \mid \mu, \sigma^2)$ :

$$p(\mu, \sigma^2 \mid y) \propto \sigma^{-2} \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu)^2\right)$$

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- Dropping the constant  $\frac{1}{\sqrt{2\pi}}$ , and using  $\prod \frac{1}{\sigma} = \sigma^{-n}$  and  $\prod \exp(a_i) = \exp(\sum a_i)$ :

$$p(\mu, \sigma^2 \mid y) \propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right)$$

# Joint posterior

$$\begin{aligned} p(\mu, \sigma^2 \mid y) &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right) \\ &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} \left[ \sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2 \right]\right) \\ &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) \end{aligned}$$

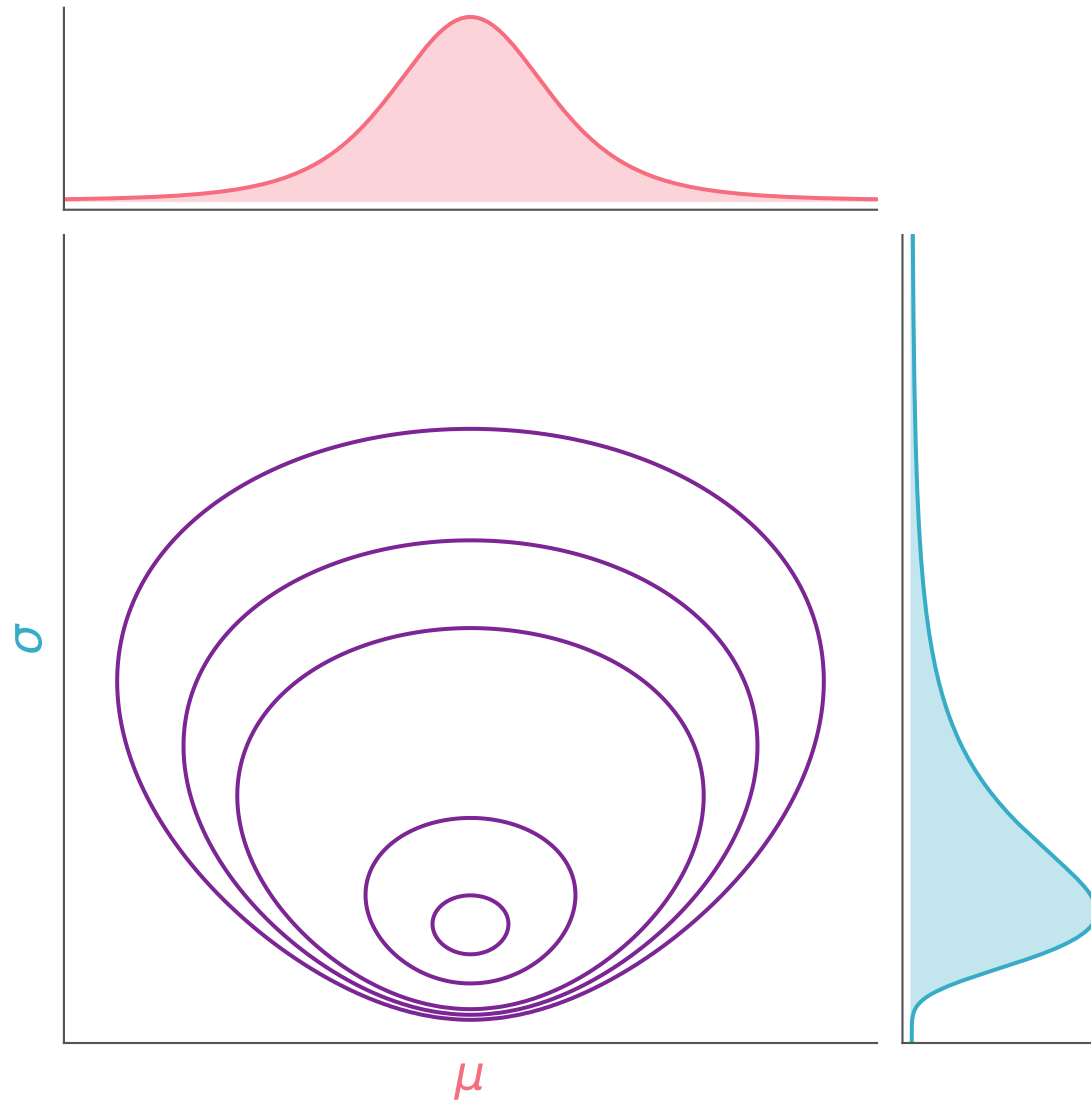
where  $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$  and  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$ .

**Step 1 → 2:** split the sum into a part that does not depend on  $\mu$  (the **within-sample** variation) plus a part that does (the **distance from the mean**). See appendix A.1.

**Step 2 → 3:** the first term is  $(n-1)$  times the sample variance  $s^2$

**Takeaway:** The only thing we need from the data are  $\bar{y}$  (mean),  $s^2$  (variance), and  $n$  (the sample size).

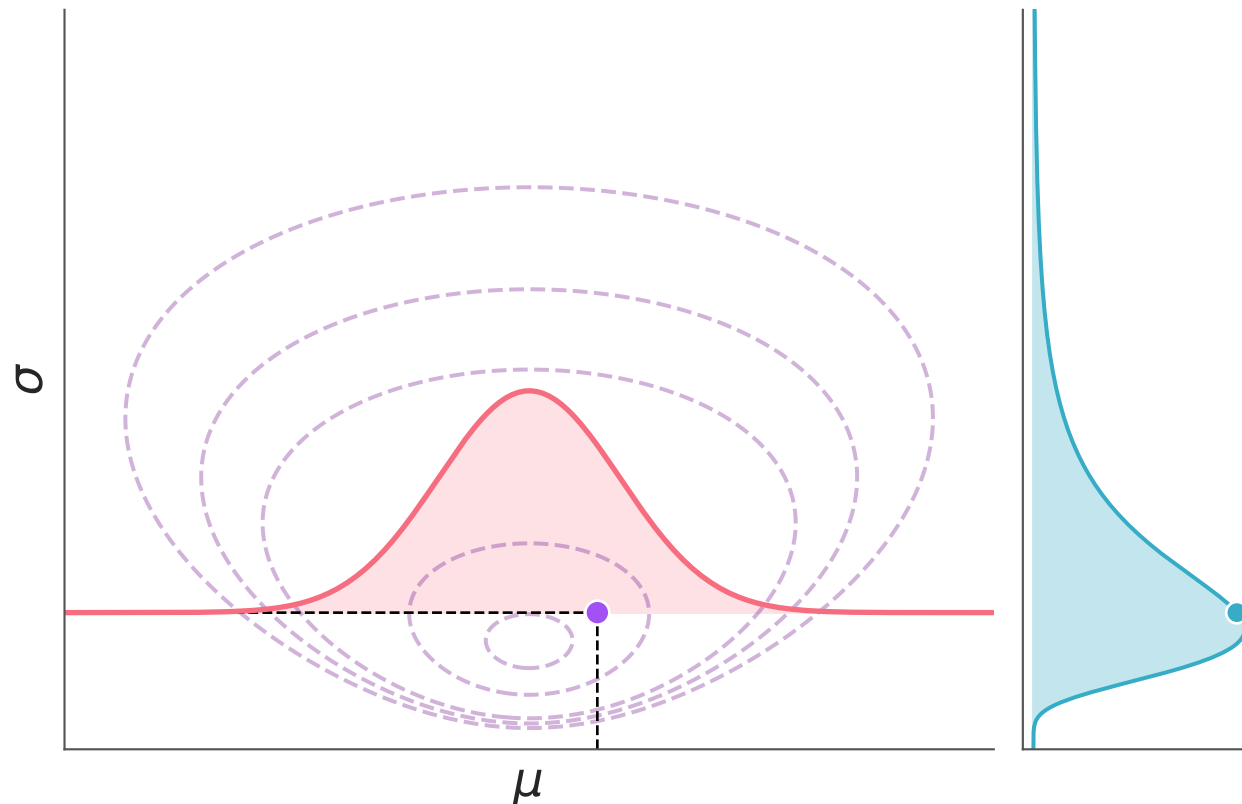
# Joint posterior with marginals



# Factorization

- Sampling from a **joint posterior** is often easier if done in steps:
  - 1. First draw from a **marginal**
  - 2. Then from a **conditional** given that draw.

$$p(\mu, \sigma^2 \mid y) = p(\mu \mid \sigma^2, y) p(\sigma^2 \mid y)$$



# Marginal posterior $p(\sigma^2 \mid y)$ , easier for $\sigma^2$ than $\sigma$

Integrate the joint posterior over the nuisance parameter  $\mu$ :

$$\begin{aligned} p(\sigma^2 \mid y) &\propto \int p(\mu, \sigma^2 \mid y) \, d\mu \\ &\propto \int \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) \, d\mu \end{aligned}$$

Split the exponential ( $e^{a+b} = e^a e^b$ ):

$$\propto \int \sigma^{-n-2} \exp\left(-(n-1)\frac{s^2}{2\sigma^2}\right) \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) \, d\mu$$

Only the second term involves  $\mu$ .

$$\propto \sigma^{-n-2} \exp\left(-(n-1)\frac{s^2}{2\sigma^2}\right) \int \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) \, d\mu$$

# Marginal posterior $p(\sigma^2 \mid y)$ (cont.)

The remaining integral is Gaussian in  $\mu$ , with

$$\mu \mid \sigma, y \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right).$$

Therefore,

$$\int \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) d\mu = \sqrt{\frac{2\pi\sigma^2}{n}} \propto \sigma.$$

Substituting back,

$$p(\sigma^2 \mid y) \propto \sigma^{-n-2} \cdot \sigma \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)$$

and hence

$$p(\sigma^2 \mid y) \propto \sigma^{-(n+1)} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right).$$

# Marginal posterior $p(\sigma^2 \mid y)$ (cont.)

Rewrite in terms of  $\sigma^2$ :

$$p(\sigma^2 \mid y) \propto (\sigma^2)^{-\frac{n+1}{2}} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right).$$

This is the kernel of a scaled inverse- $\chi^2$  distribution:

$$\sigma^2 \mid y \sim \text{Inv} - \chi^2(n-1, s^2)$$

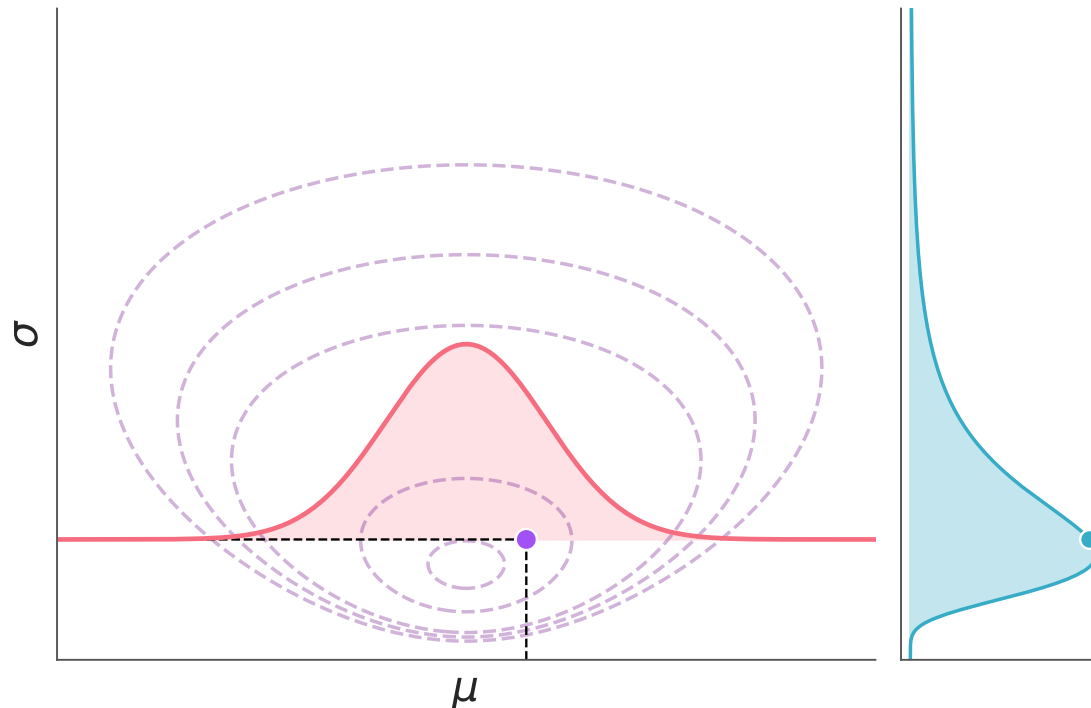
# Factorization

- Sampling from a **joint posterior** is often easier if done in steps:
  - 1. Sample for the marginal:

$$p(\sigma^2 \mid y) = \text{Inv-}\chi^2(n - 1, s^2)$$

- 2. Given that draw, then from a **conditional** given that draw.

$$p(\mu \mid \sigma^2, y) = N\left(\bar{y}, \frac{\sigma^2}{n}\right)$$



# Factorization

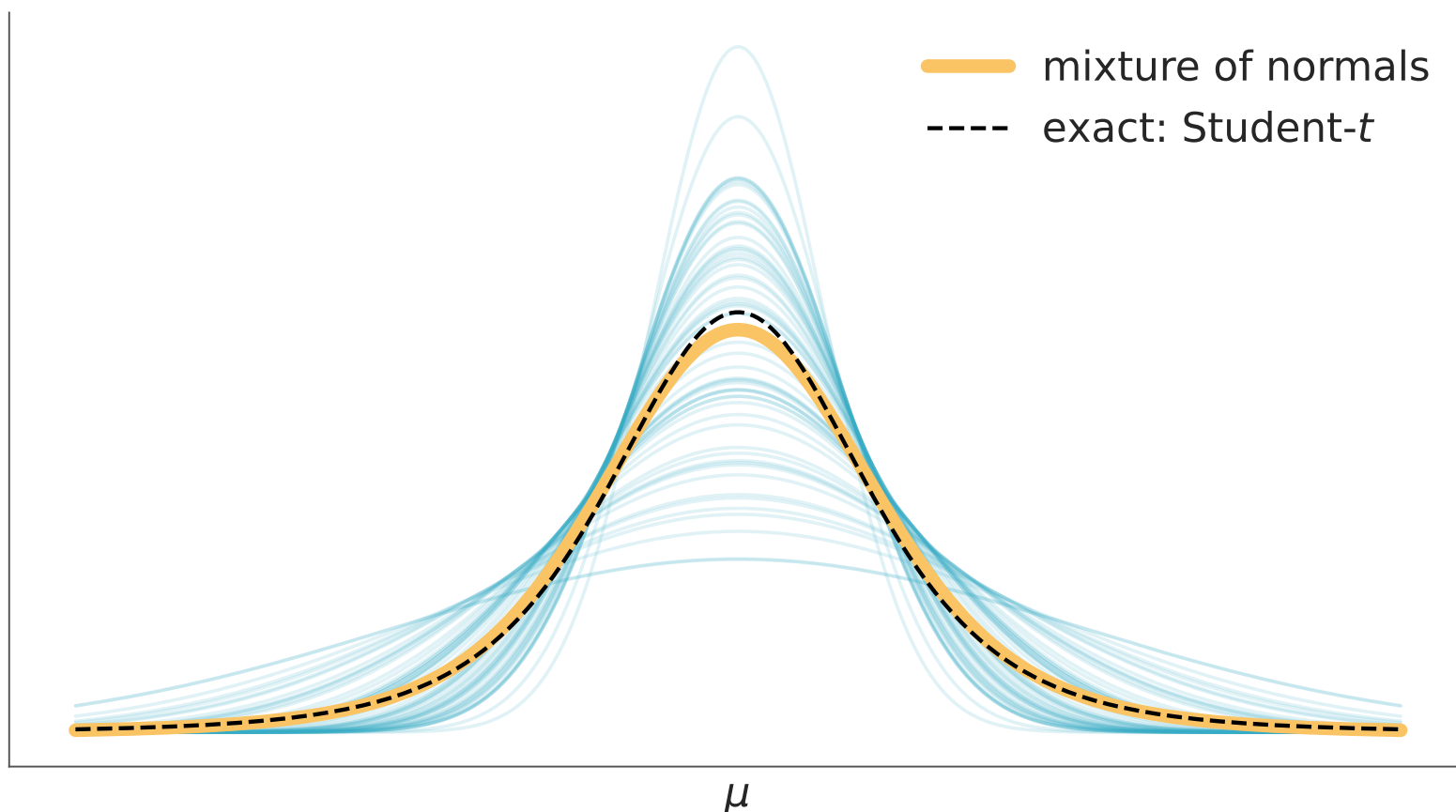
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# Factorization

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- Averaging over all these conditional normals approximates the marginal posterior for  $\mu$  (**a mixture of normals**):
- This mixture is in fact a **Student- $t$  distribution** (See appendix A.2)



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$$\tilde{y}^s \sim p(\tilde{y} \mid \mu^s, \sigma^s)$$

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- Step 2: draw a new observation from the data model

$$\tilde{y}^s \sim p(\tilde{y} \mid \mu^s, \sigma^s)$$

- Step 3: repeat steps 1 and 2 many times to approximate the predictive distribution

# Posterior predictive distribution (Normal with known variance)

We integrate the data model over the conditional posterior for  $\mu$ :

$$p(\tilde{y} \mid \sigma^2, y) = \int p(\tilde{y} \mid \mu, \sigma^2) p(\mu \mid \sigma^2, y) d\mu$$

$$p(\tilde{y} \mid \sigma^2, y) = \int N(\mu, \sigma^2) N\left(\bar{y}, \frac{\sigma^2}{n}\right) d\mu$$

The integral of a normal against a normal is again normal:

$$= N\left(\bar{y}, \sigma^2 + \frac{\sigma^2}{n}\right) = N\left(\bar{y}, \left(1 + \frac{1}{n}\right)\sigma^2\right)$$

The *extra*  $\frac{\sigma^2}{n}$  comes from not knowing  $\mu$  exactly.

# Posterior predictive distribution

In practice  $\sigma^2$  is **not** known, so we also integrate over its posterior uncertainty:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \sigma^2, y) p(\sigma^2 \mid y) d\sigma^2$$

Integrating over the uncertainty in  $\sigma^2$  gives a Student- $t$ , with  $s^2$  in place of  $\sigma^2$ :

$$p(\tilde{y} \mid y) = t_{n-1} \left( \bar{y}, \left( 1 + \frac{1}{n} \right) s^2 \right)$$

Compared with  $p(\mu \mid y) = t_{n-1} \left( \bar{y}, \frac{s^2}{n} \right)$ :

$$\frac{s^2}{n} \quad \text{vs.} \quad \left( 1 + \frac{1}{n} \right) s^2 = \frac{s^2}{n} + s^2$$

Predicting a new observation is less certain than estimating  $\mu$ : it includes parameter uncertainty + observation noise.

# Normal - conjugate prior

- There are conjugate priors for the normal model
  - Check Section 3.3 in BDA3 for details.
  - Wikipedia has a list of [conjugate priors](#) for various models
- Useful sometimes when speed and analytical convenience (or insight) are important.
- With modern methods like MCMC, conjugate priors are less critical for computation.

# Comparison of means of two normals

- The difference of two normally distributed variables is normally distributed
- The difference of two  $t$  distributed variables with different variances and degrees of freedom doesn't have a closed form
  - but easy to sample from the two distributions, and obtain draws of the differences

$$\text{if } \mu_1^s \sim p(\mu_1 \mid y_1)$$

$$\mu_2^s \sim p(\mu_2 \mid y_2)$$

$$\delta^s = \mu_1^s - \mu_2^s$$

$$\text{then } \delta^s \sim p(\delta \mid y_1, y_2)$$

- This is related to Monte Carlo method that will be discussed in the next lecture.

# Normal linear regression

- We allow  $\mu$  to depend on predictors  $x_i$  through a linear model:

$$y_i \sim N(\alpha + \beta x_i, \sigma^2), \quad i = 1, \dots, N$$

- For some case we can have analytical solutions for the posterior
- For the general case with arbitrary priors and unknown  $\sigma^2$ , numerical methods are needed (discussed later in the course).
- more about regression in BDA3 Chapter 14 (not required for the course) and [Regression and Other Stories](#) book

# Generalized linear model (GLM)

- We generalized regression models to allow for non-normal observation models.

$$y_i \sim f(g^{-1}(\eta_i), \varphi)$$

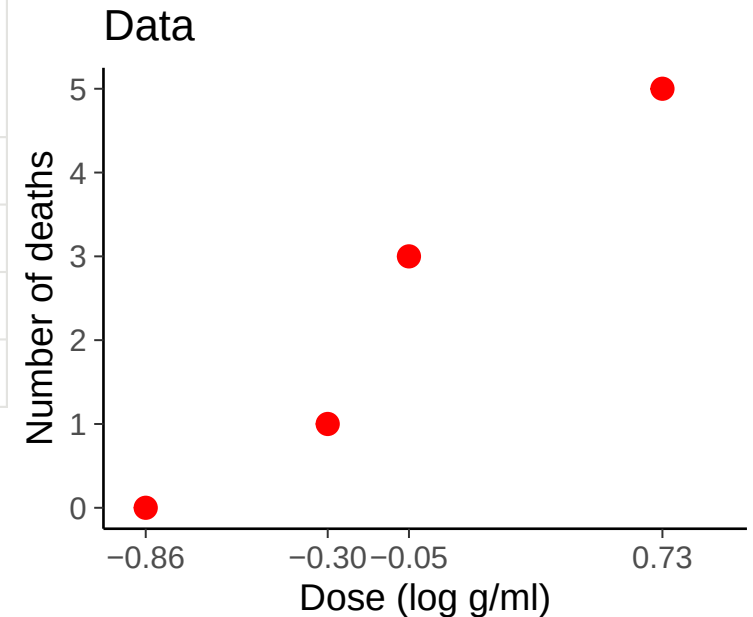
$$\eta_i = \alpha + \beta x_i$$

- $\eta$ : linear predictor term  $\eta$
- $g^{-1}$ : inverse link function
- $f$ : observation model
- $\varphi$ : other parameters of the observation model (e.g., variance for normal)

More in BDA3 Chapter 16 and [Regression and Other Stories](#) book

# Bioassay

| Dose $x_i$ (log g/ml) | Number of animals $n_i$ | Number of deaths $y_i$ |
|-----------------------|-------------------------|------------------------|
| -0.86                 | 5                       | 0                      |
| -0.30                 | 5                       | 1                      |
| -0.05                 | 5                       | 3                      |
| 0.73                  | 5                       | 5                      |



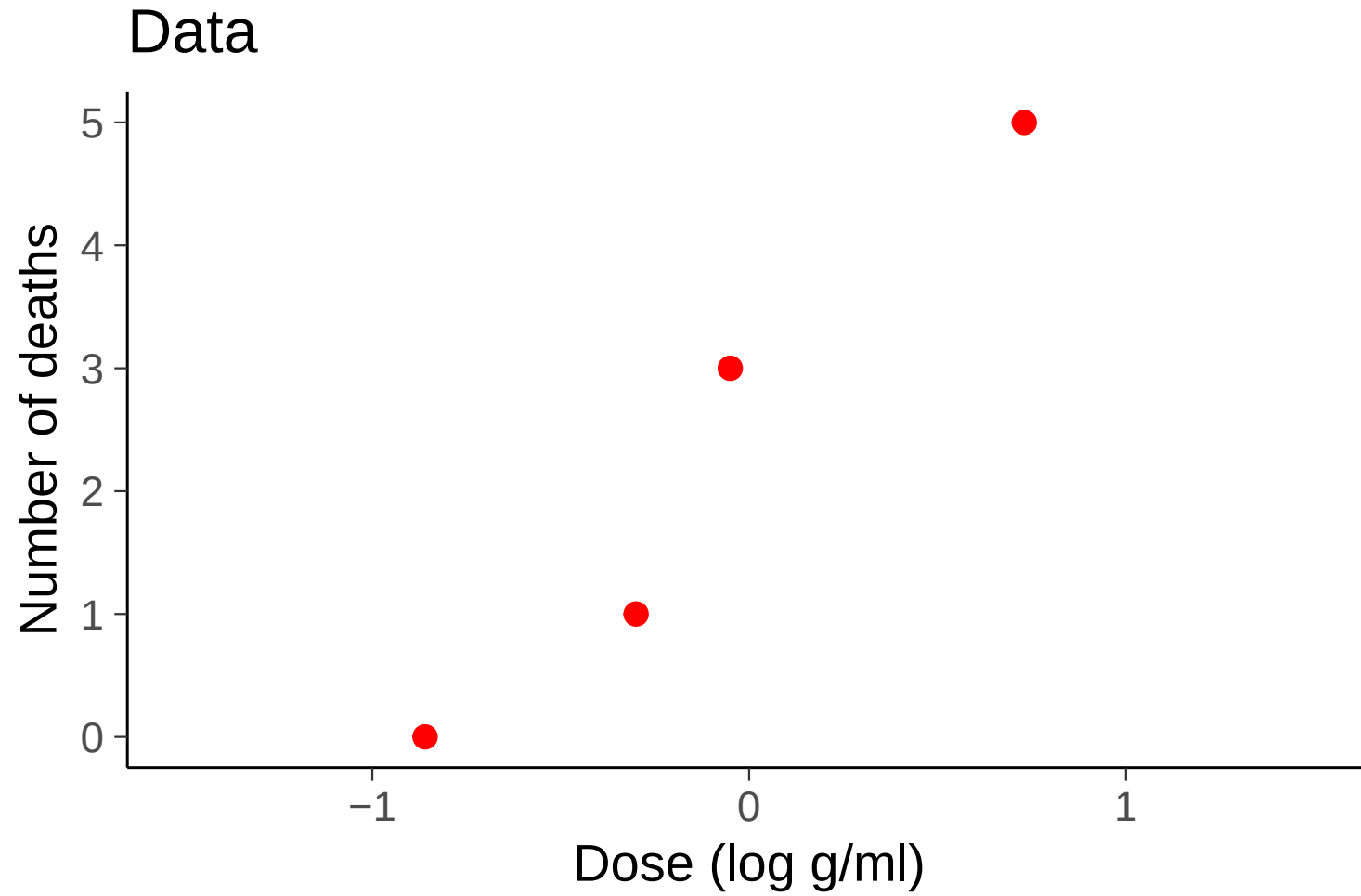
Find out lethal dose 50% (LD50)

- used to classify how hazardous chemical is
- 1984 EEC directive has 4 levels (see the chapter notes)

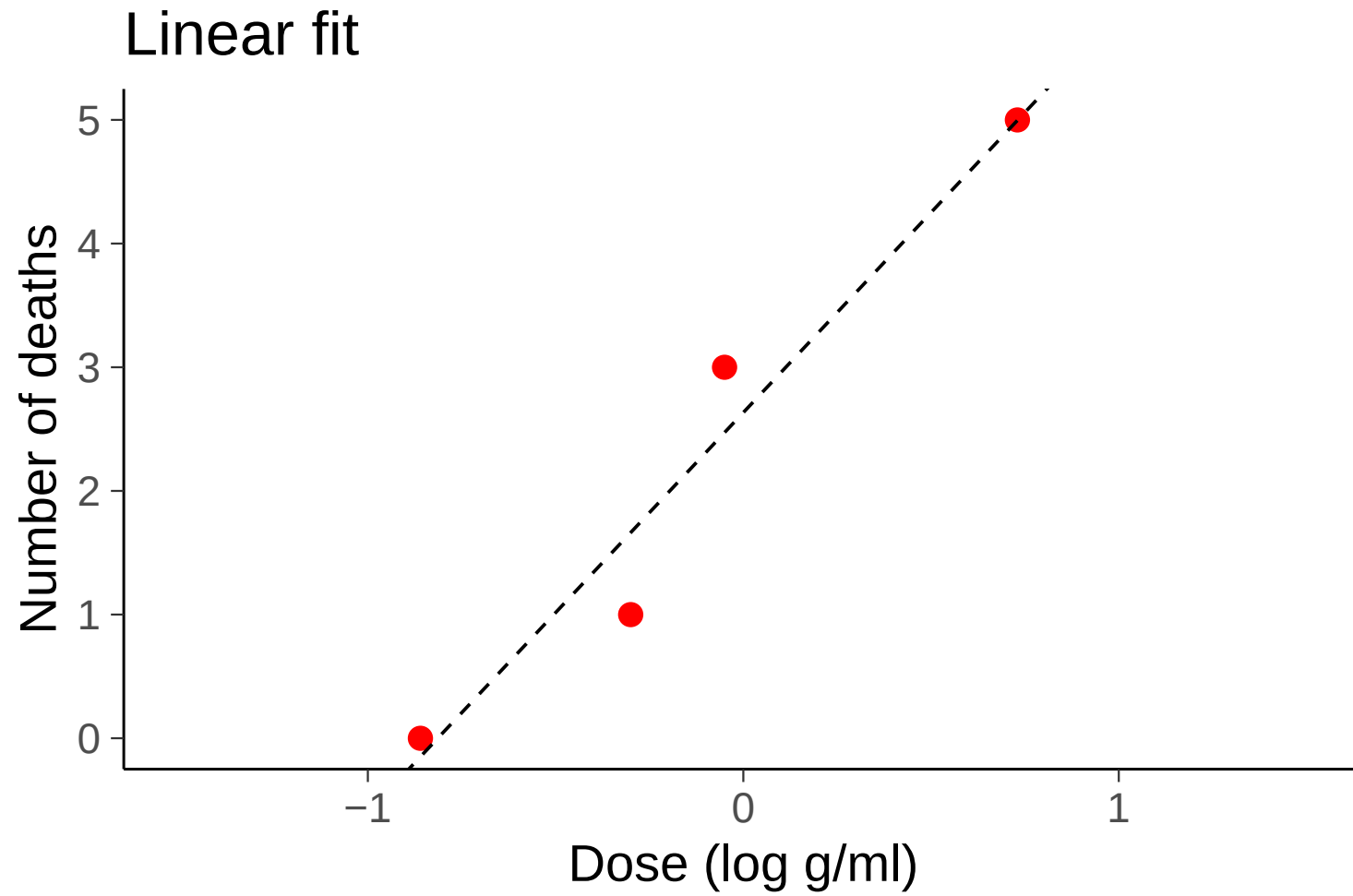
Bayesian methods help to

- reduce the number of animals needed
- easy to make sequential experiment and stop as soon as desired accuracy is obtained

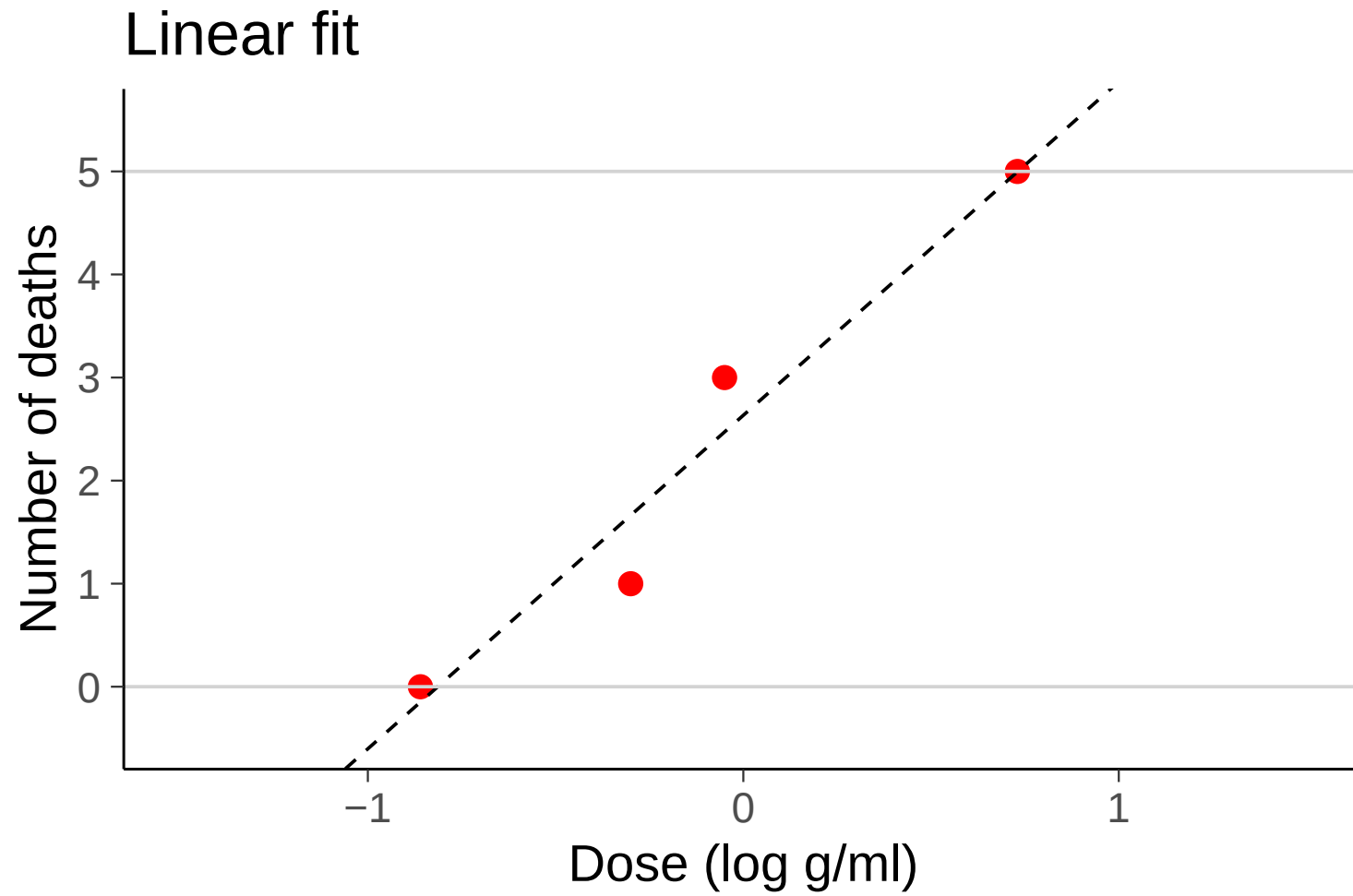
# Bioassay



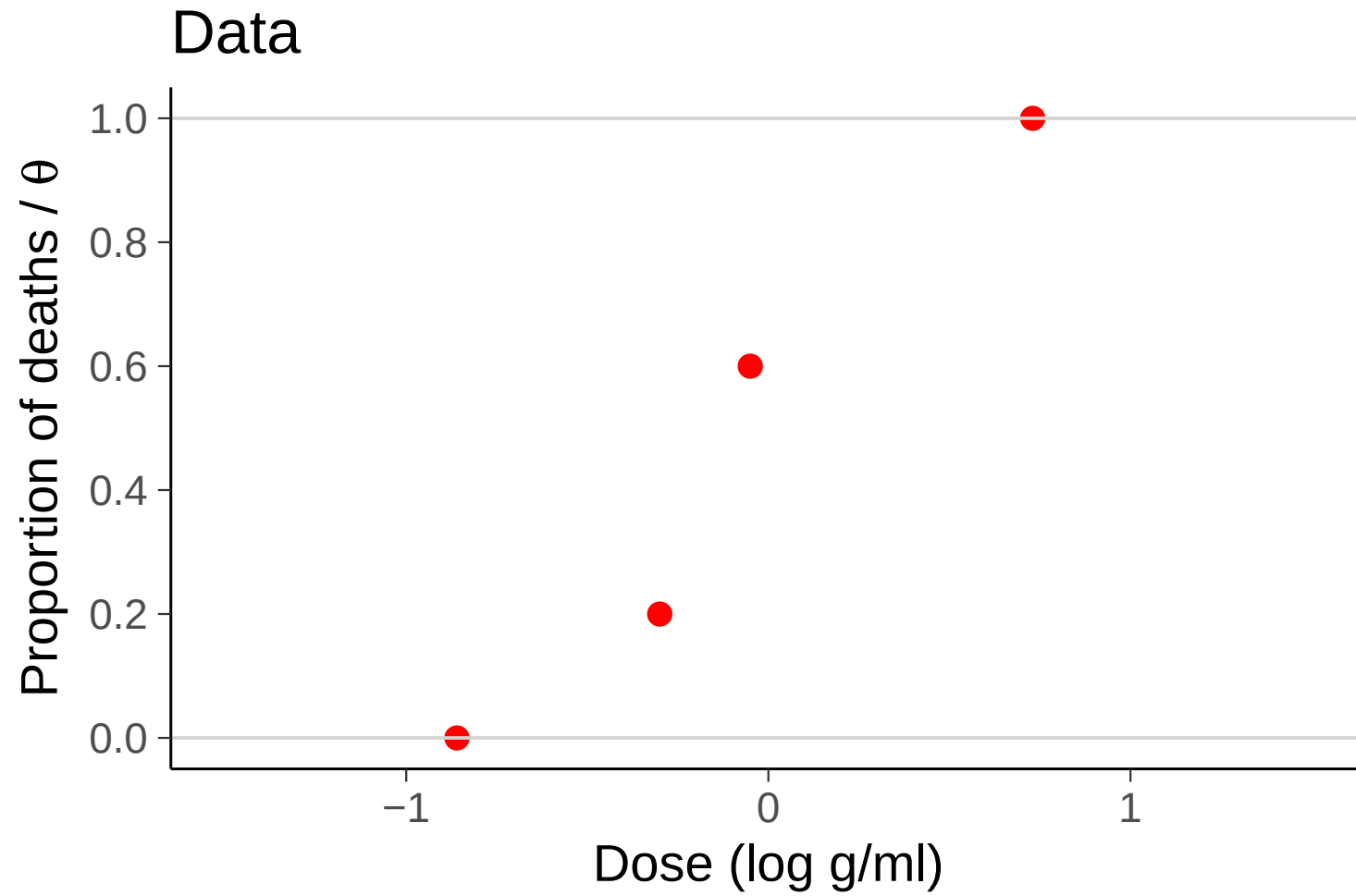
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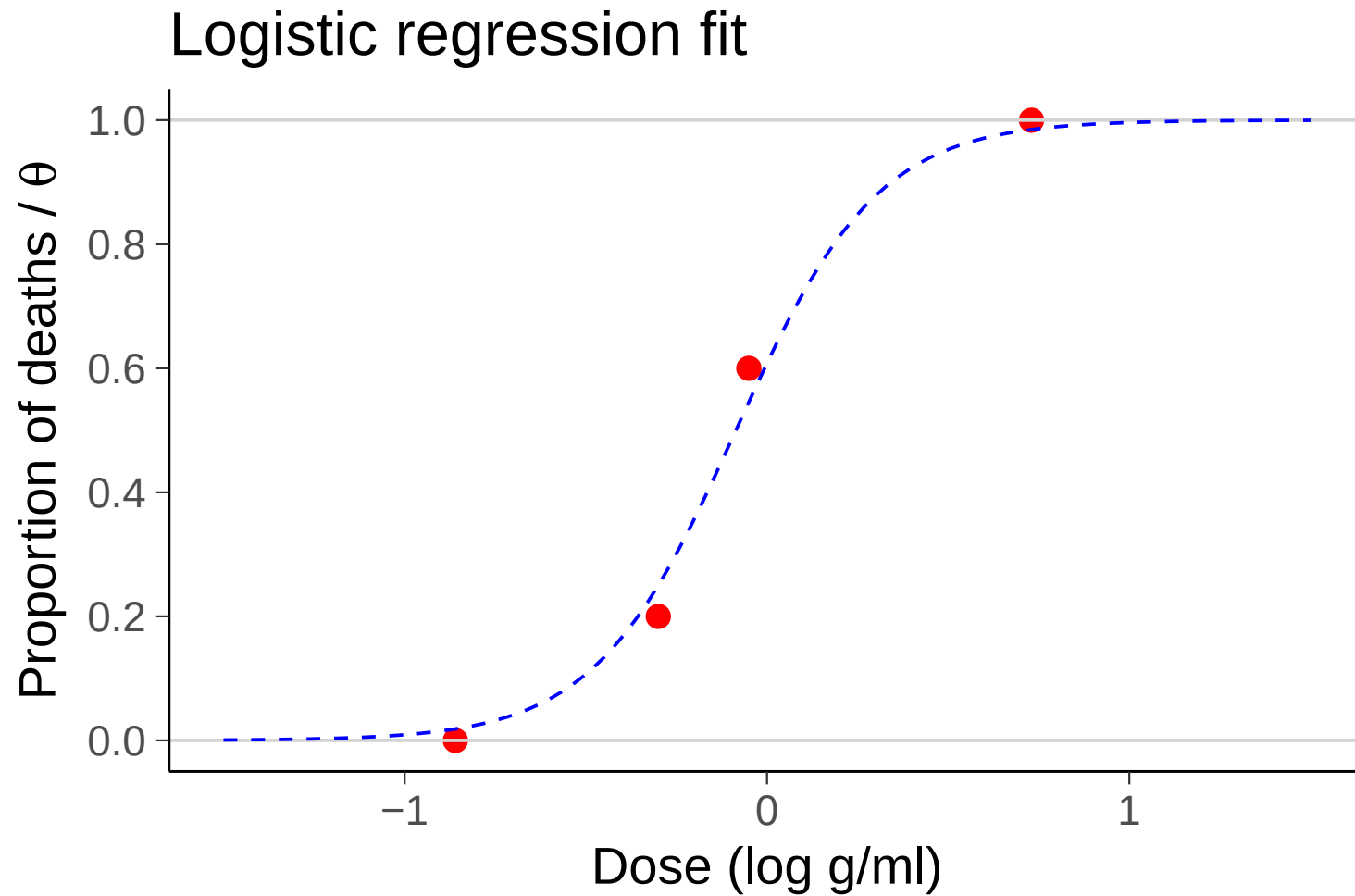
# Bioassay



# Bioassay



# Bioassay



Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i)$$

# Bioassay (GLM for binomial data)

- Generalized linear model

$$y_i \sim f(g^{-1}(\eta_i), \varphi)$$

$$\eta_i = \alpha + \beta x_i$$

- GLM for binomial data
  - $f$  is the Binomial model
  - $g^{-1}$  is the logistic function
  - $g$  is the logit function
  - $\varphi$  is not needed for the Binomial model

# Bioassay

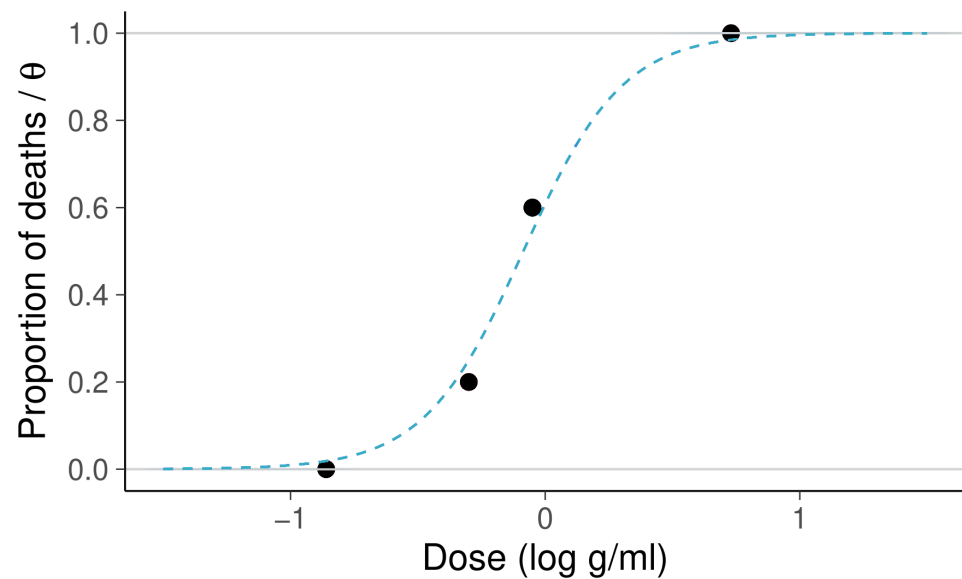
$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i)$$

$$\text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right)$$

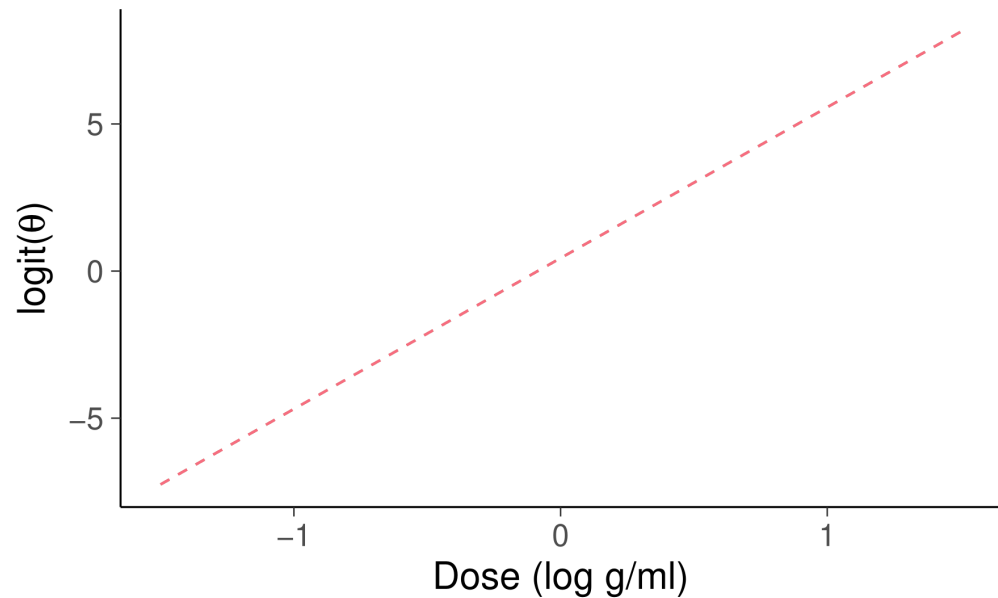
$$= \alpha + \beta x_i$$

$$\theta_i = \frac{1}{1 + \exp(-(\alpha + \beta x_i))}$$

Logistic regression fit

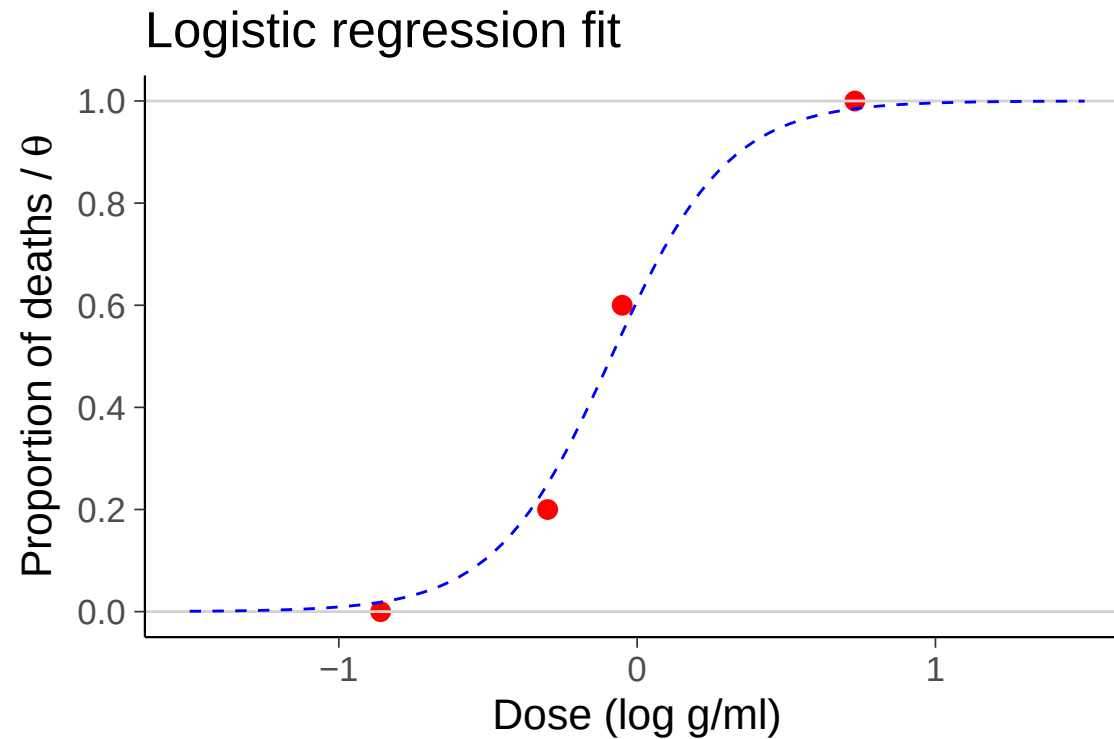


Logistic regression in latent space



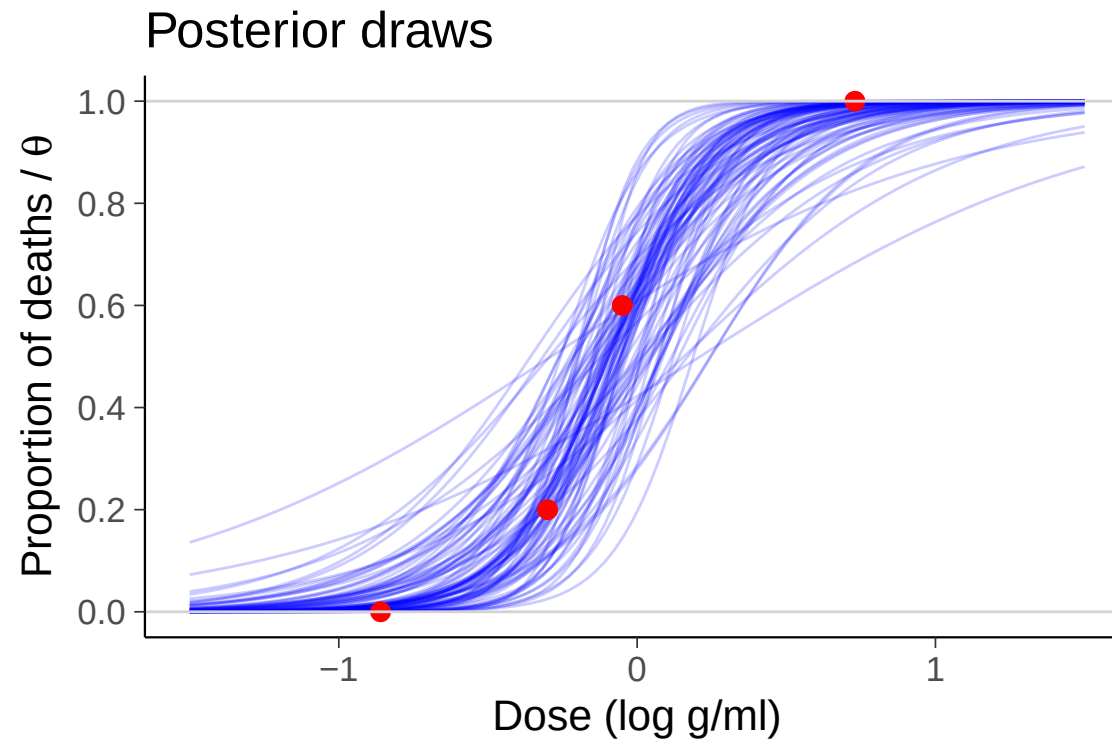
# Bioassay

- One curve, the posterior mean of the dose-response relationship

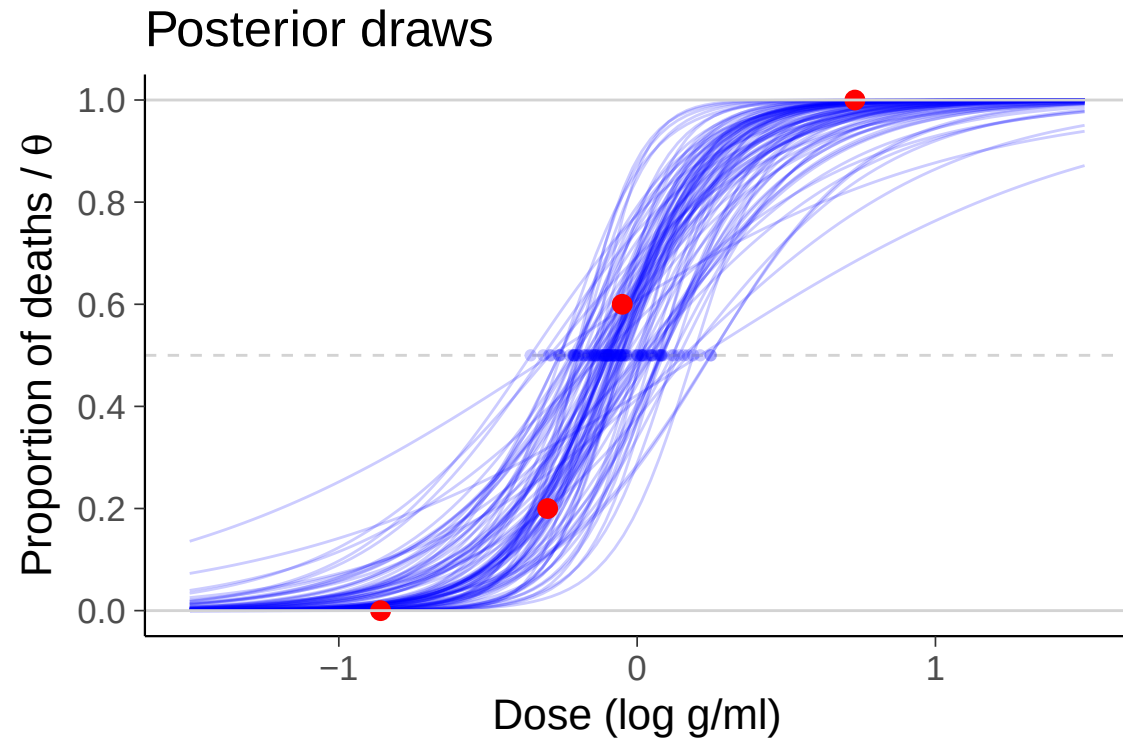


# Bioassay

- Many curves, representing samples from the posterior distribution of the dose-response relationship



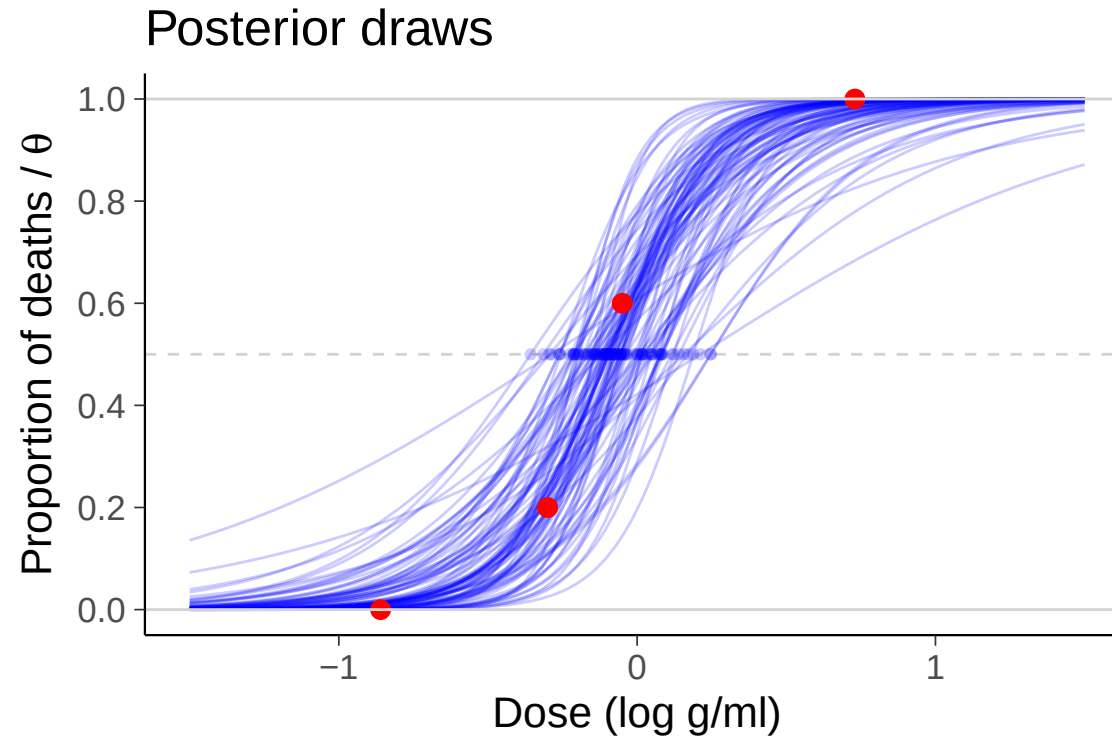
# Bioassay



$$\text{LD50: } E[\theta] = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \quad \Rightarrow \quad x_{\text{LD50}} = -\frac{\alpha}{\beta}$$

# Bioassay

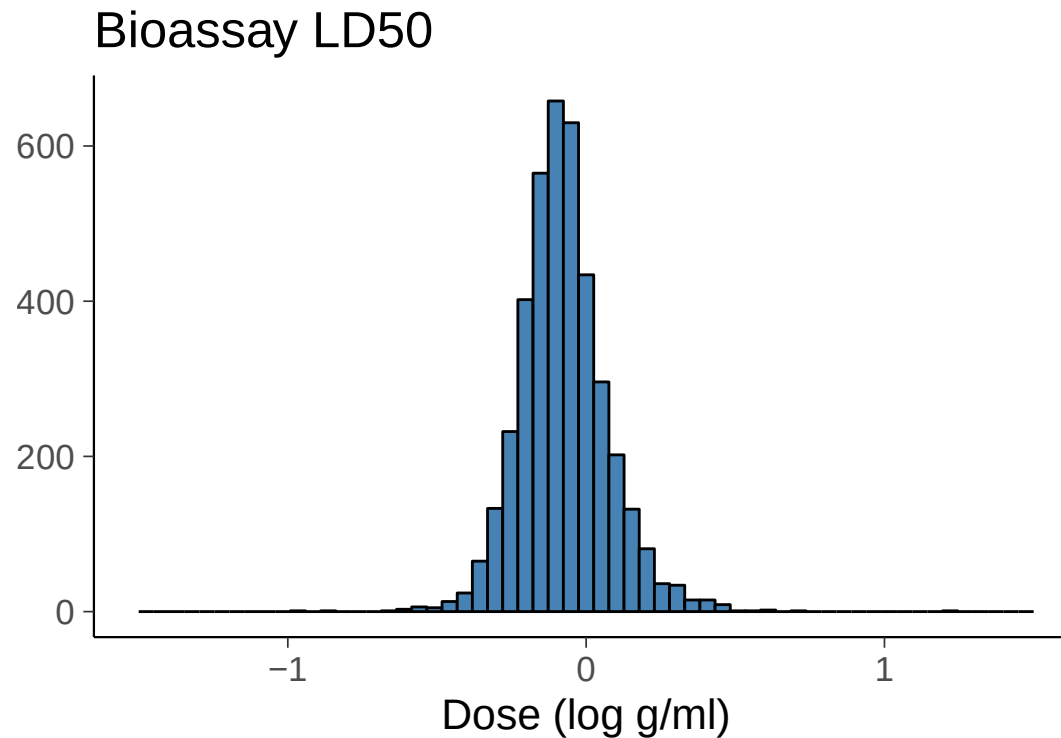
- Posterior distribution of the LD50



$$\text{LD50: } E[\theta] = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \Rightarrow x_{\text{LD50}} = -\frac{\alpha}{\beta} \Rightarrow x_{\text{LD50}}^s = -\frac{\alpha^s}{\beta^s}$$

# Bioassay

- Histogram of the posterior distribution of the LD50



$$\text{LD50: } E[\theta] = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \Rightarrow x_{\text{LD50}} = -\frac{\alpha}{\beta} \Rightarrow x_{\text{LD50}}^s = -\frac{\alpha^s}{\beta^s}$$

# Bioassay posterior

- Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i)$$

- Link function

$$\text{logit}(\theta_i) = \alpha + \beta x_i$$

- Likelihood

$$p(y_i \mid \alpha, \beta, n_i, x_i) \propto \theta_i^{y_i} [1 - \theta_i]^{n_i - y_i}$$

$$p(y_i \mid \alpha, \beta, n_i, x_i) \propto [\text{logit}^{-1}(\alpha + \beta x_i)]^{y_i} [1 - \text{logit}^{-1}(\alpha + \beta x_i)]^{n_i - y_i}$$

Posterior (with uniform prior on  $\alpha, \beta$ )

$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{i=1}^n p(y_i \mid \alpha, \beta, n_i, x_i)$$

# Bioassay - grid approximation

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- Choose a grid of values for  $\alpha$  and  $\beta$ :  $\alpha_1, \dots, \alpha_M$  and  $\beta_1, \dots, \beta_N$

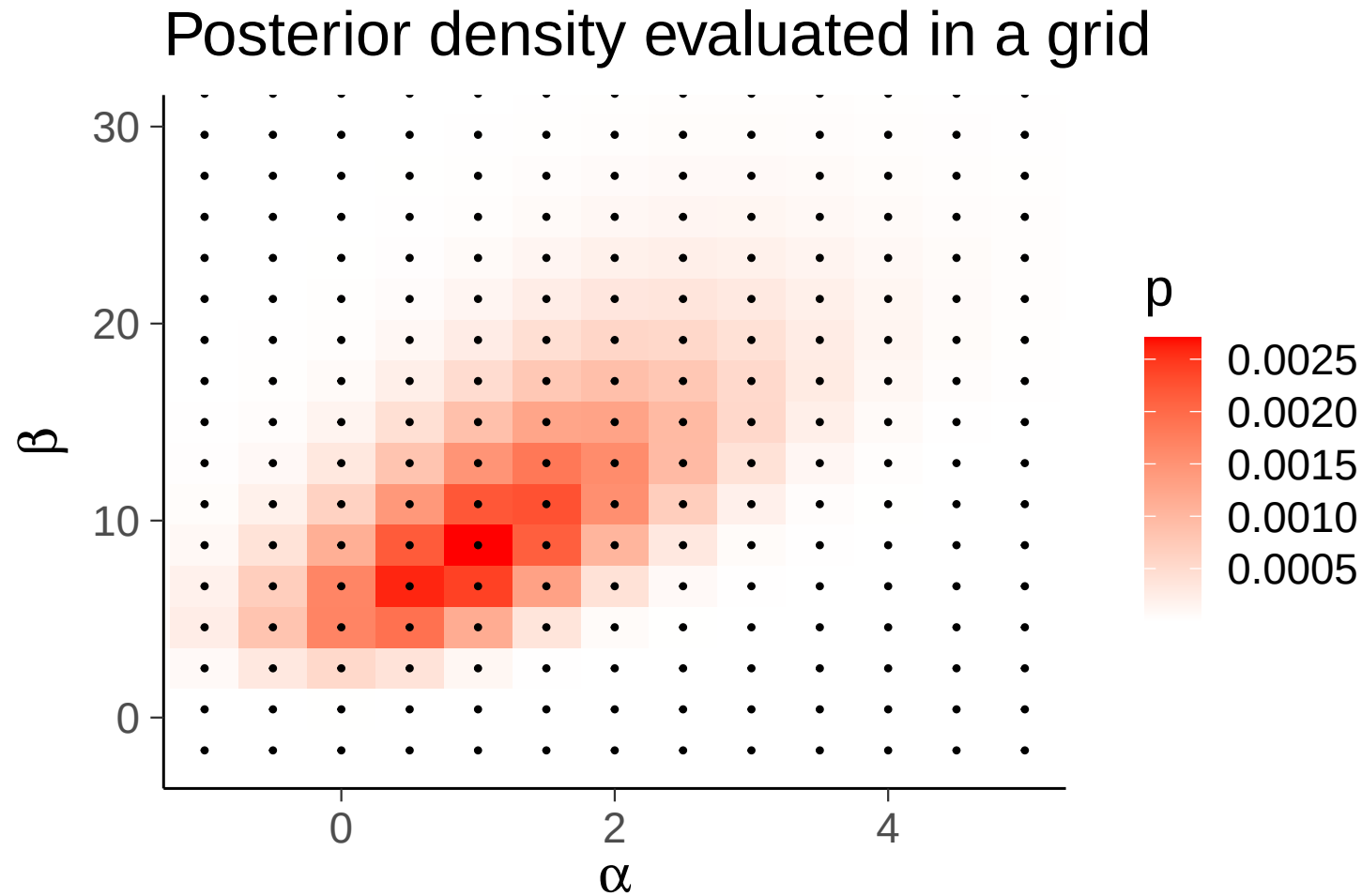
# Bioassay - grid approximation

- The posterior  $p(\alpha, \beta \mid y, n, x)$  has no closed-form
- But we can evaluate the posterior at any given point in the parameter space.
- Choose a grid of values for  $\alpha$  and  $\beta$ :  $\alpha_1, \dots, \alpha_M$  and  $\beta_1, \dots, \beta_N$
- Evaluate the unnormalized posterior at every grid point  $(\alpha_j, \beta_k)$

$$q_{j,k} = p(\alpha_j, \beta_k \mid y, n, x) \propto p(\alpha_j, \beta_k) \prod_{i=1}^n p(y_i \mid \alpha_j, \beta_k, n_i, x_i)$$

# Bioassay - grid approximation

- Density times cell area gives probability mass in each cell



# Bioassay - grid approximation

## Normalization

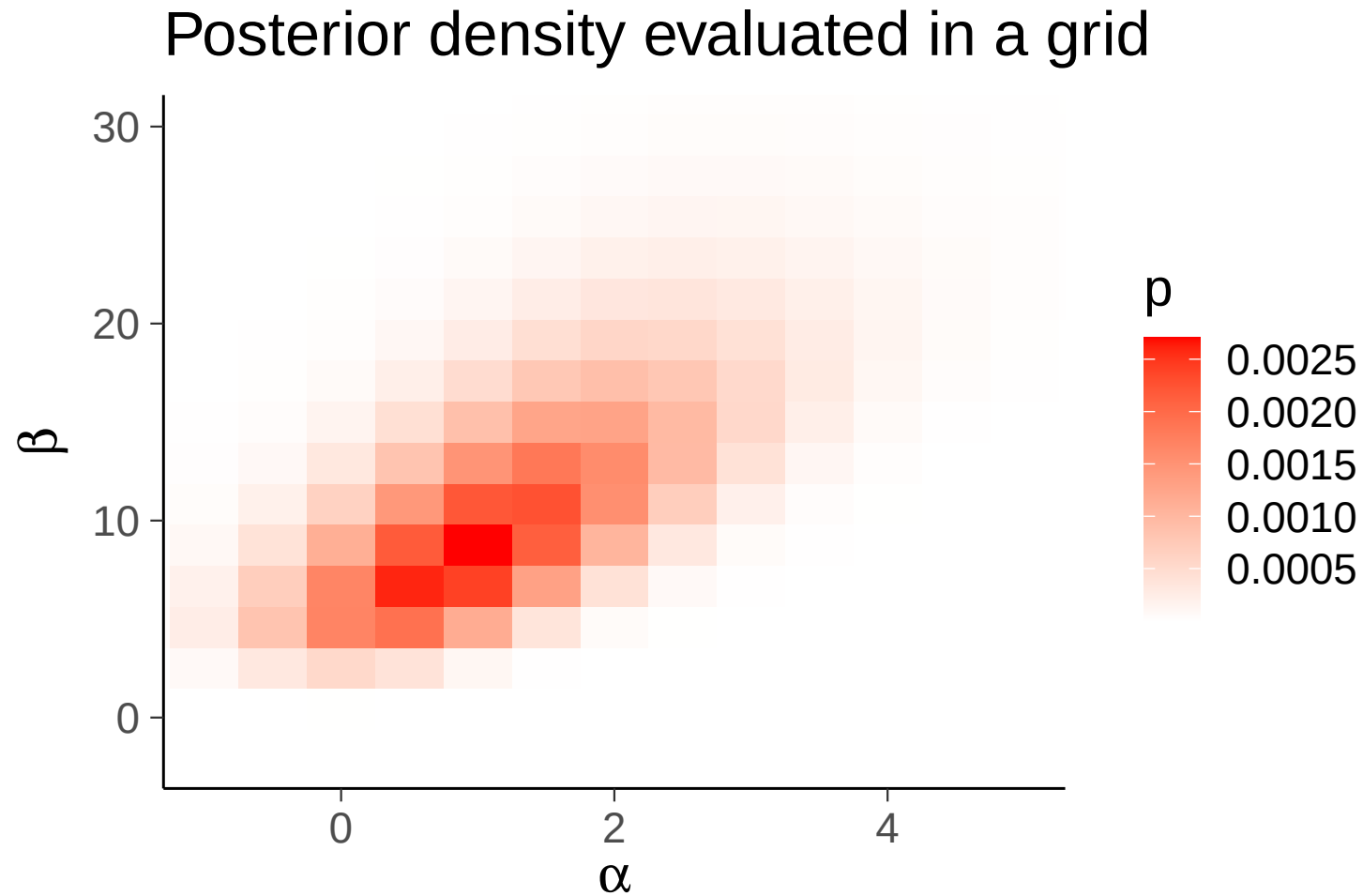
- Since we only know the posterior up to a constant, we normalize over the grid:

$$p(\alpha_j, \beta_k \mid y, n, x) \approx \frac{q_{j,k}}{\sum_{j,k} q_{j,k}}$$

- This turns the grid of unnormalized values into a proper discrete probability distribution that sums to 1.

# Bioassay - grid

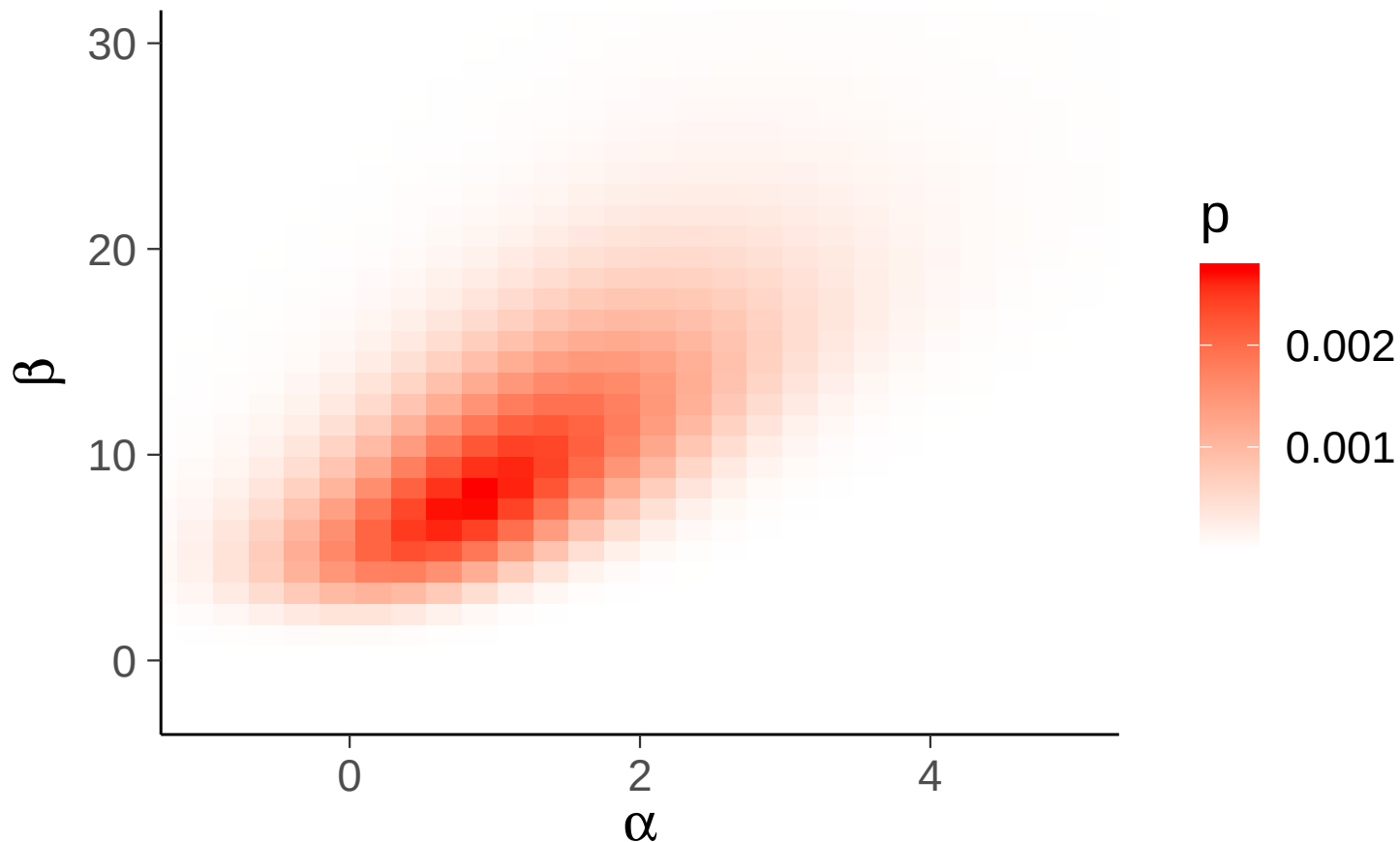
Density evaluated in a coarse grid



# Bioassay - grid

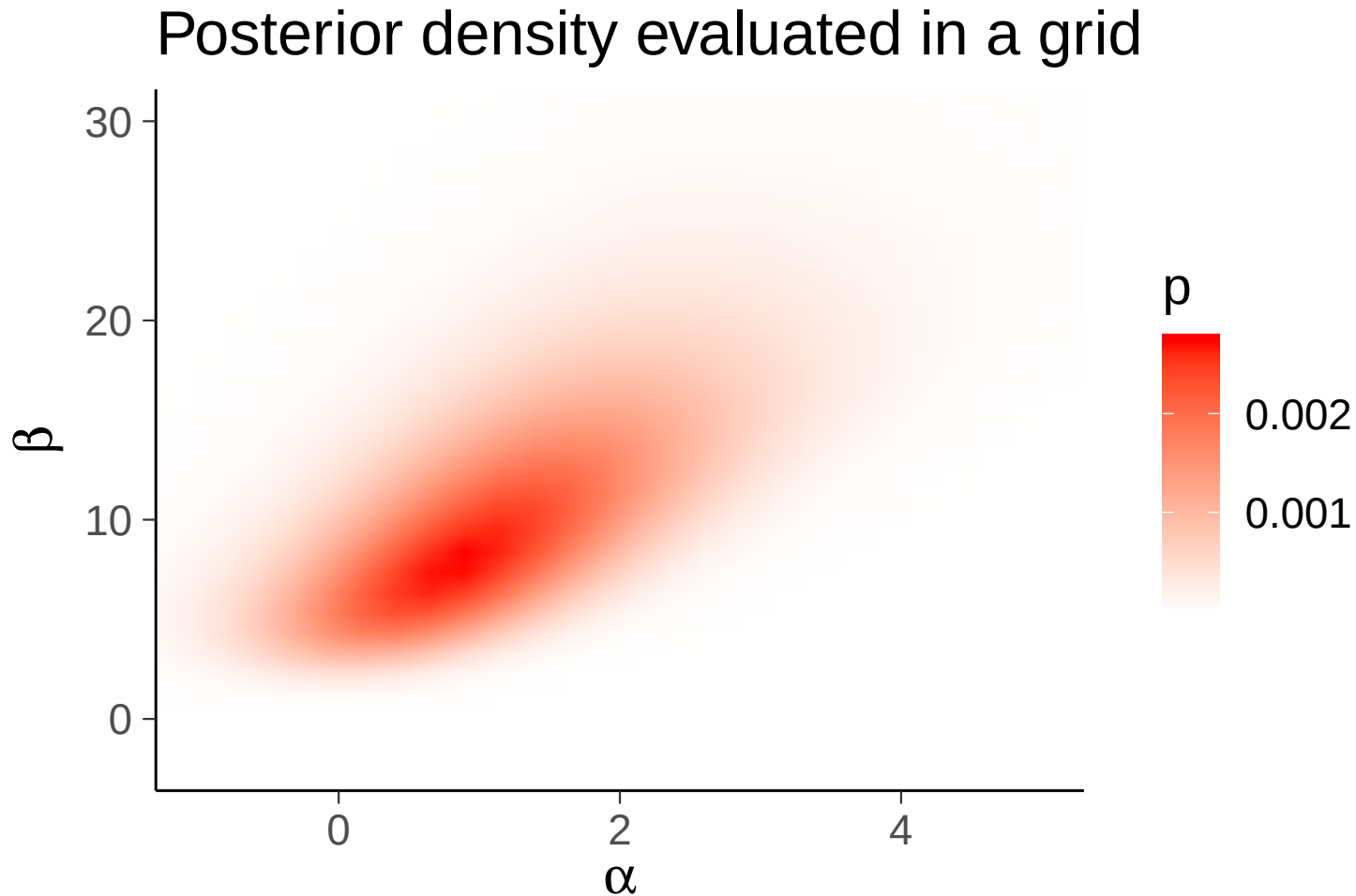
- Density evaluated in a finer grid
  - The finer the grid, the more accurate the approximation becomes
  - But cost can grow fast, in particular as the number of parameters increases

Posterior density evaluated in a grid



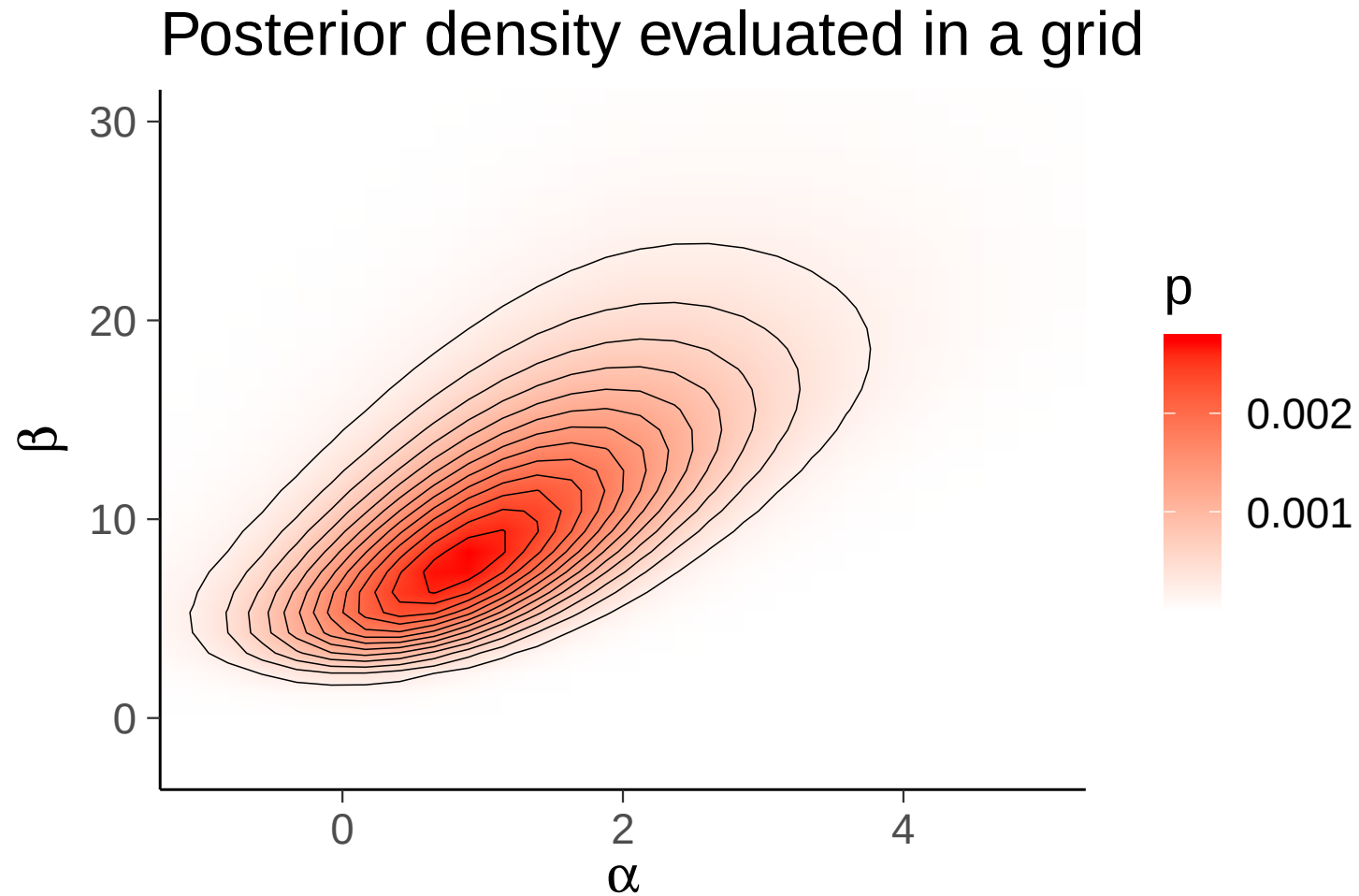
# Bioassay - grid

- For an smoother visualization, we may want to interpolate



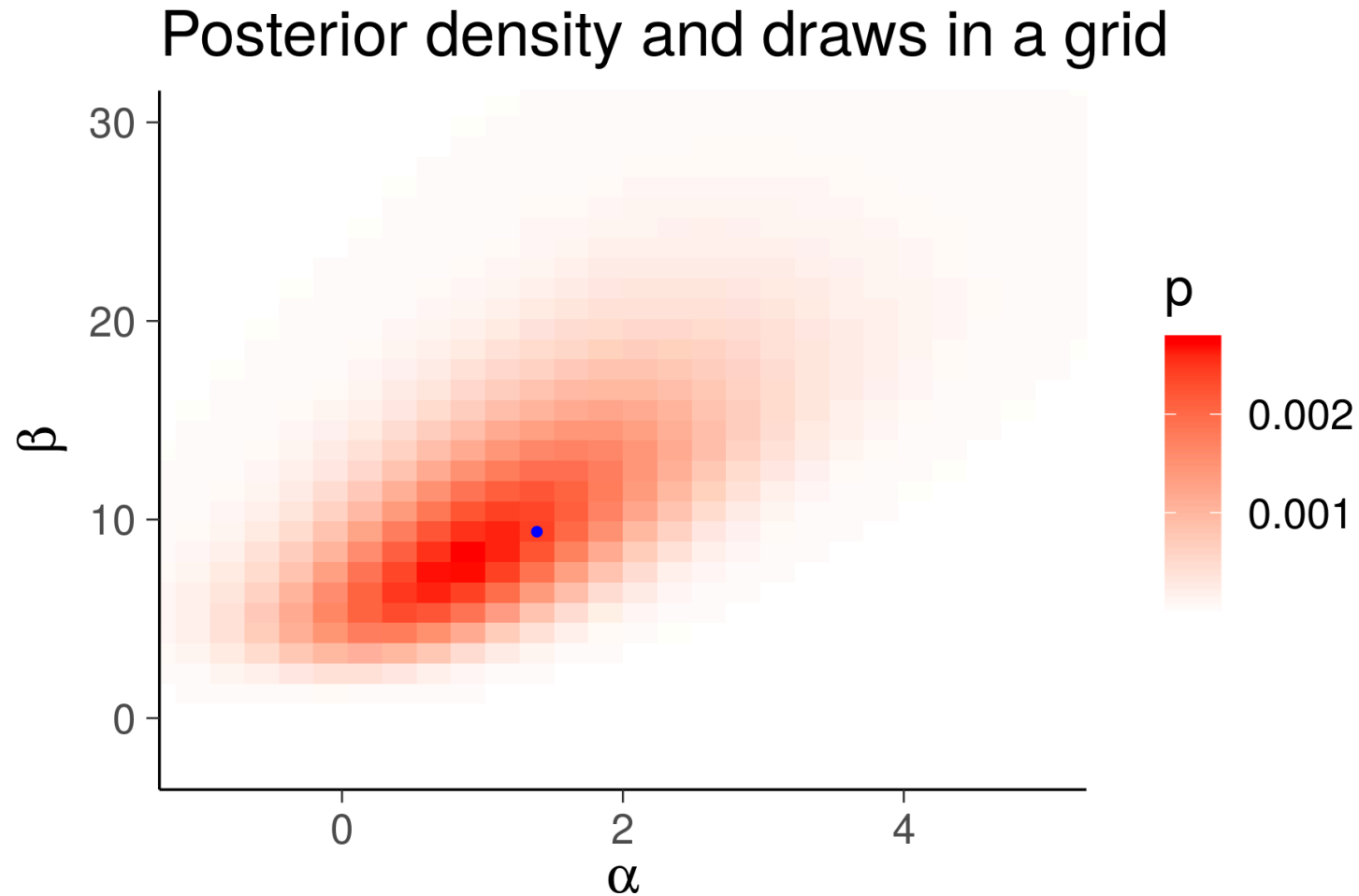
# Bioassay - grid

- And maybe also compute contour lines



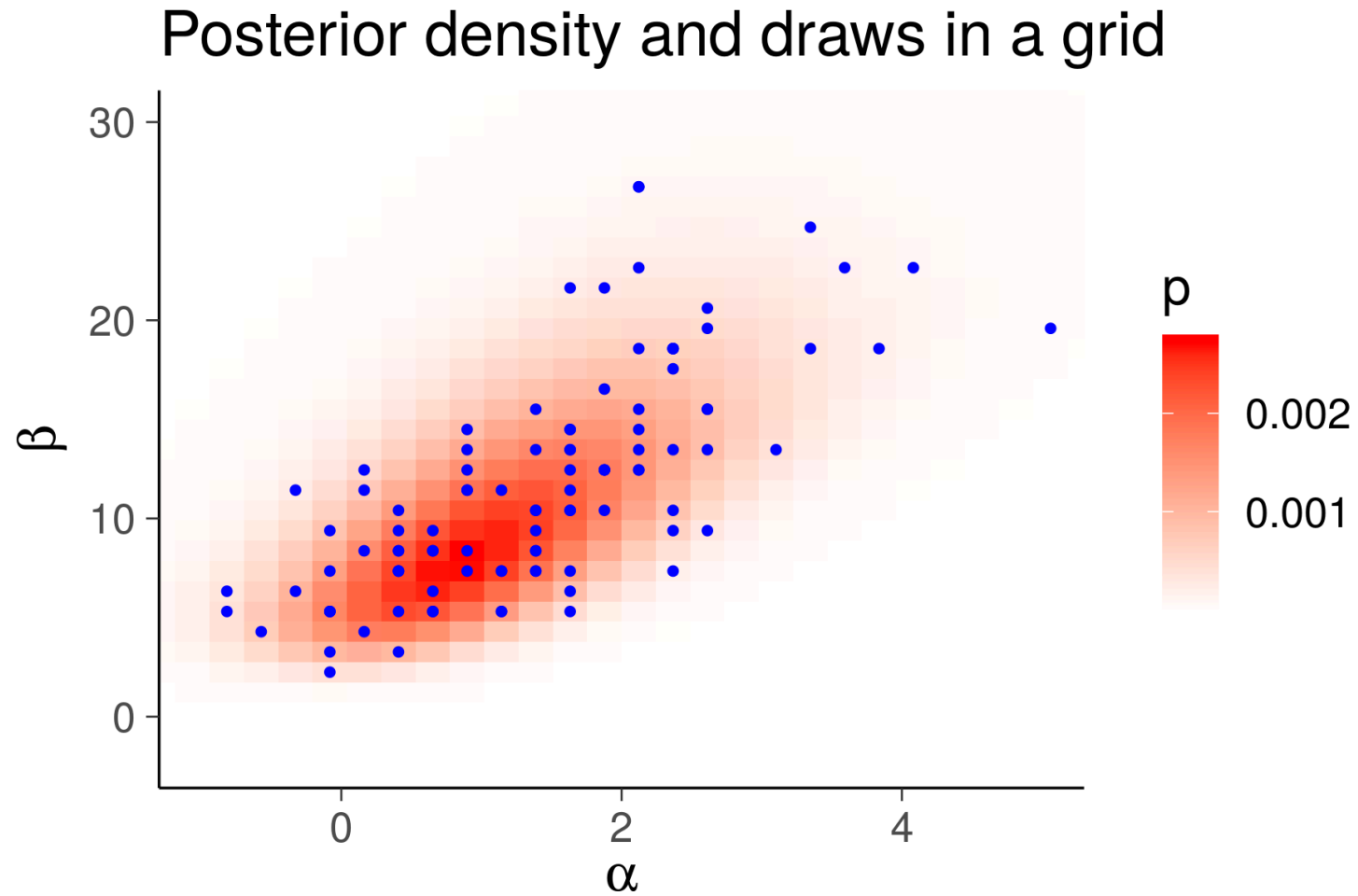
# Bioassay - grid

- We can sample based on the grid cell probabilities



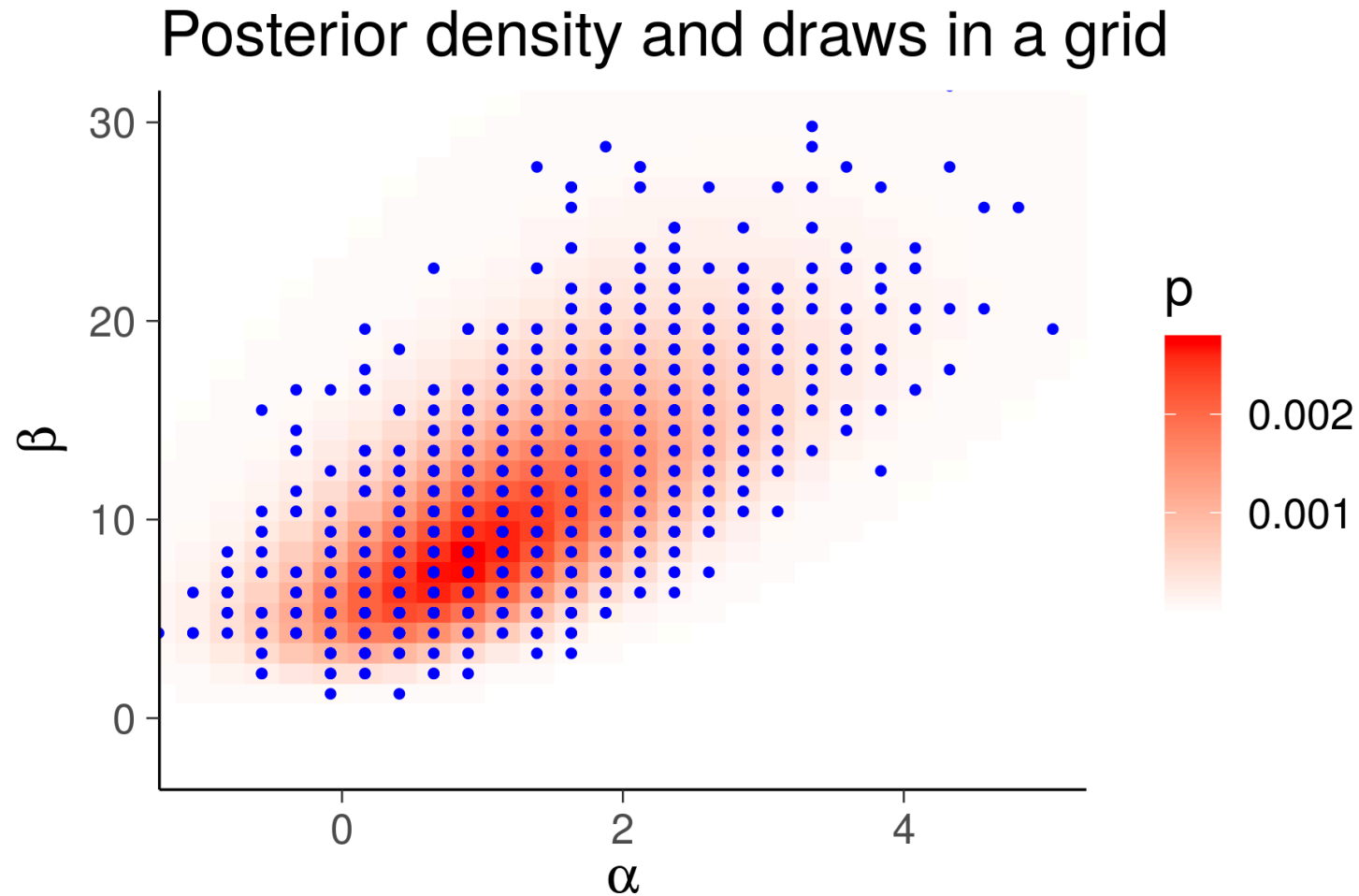
# Bioassay - grid

- More samples



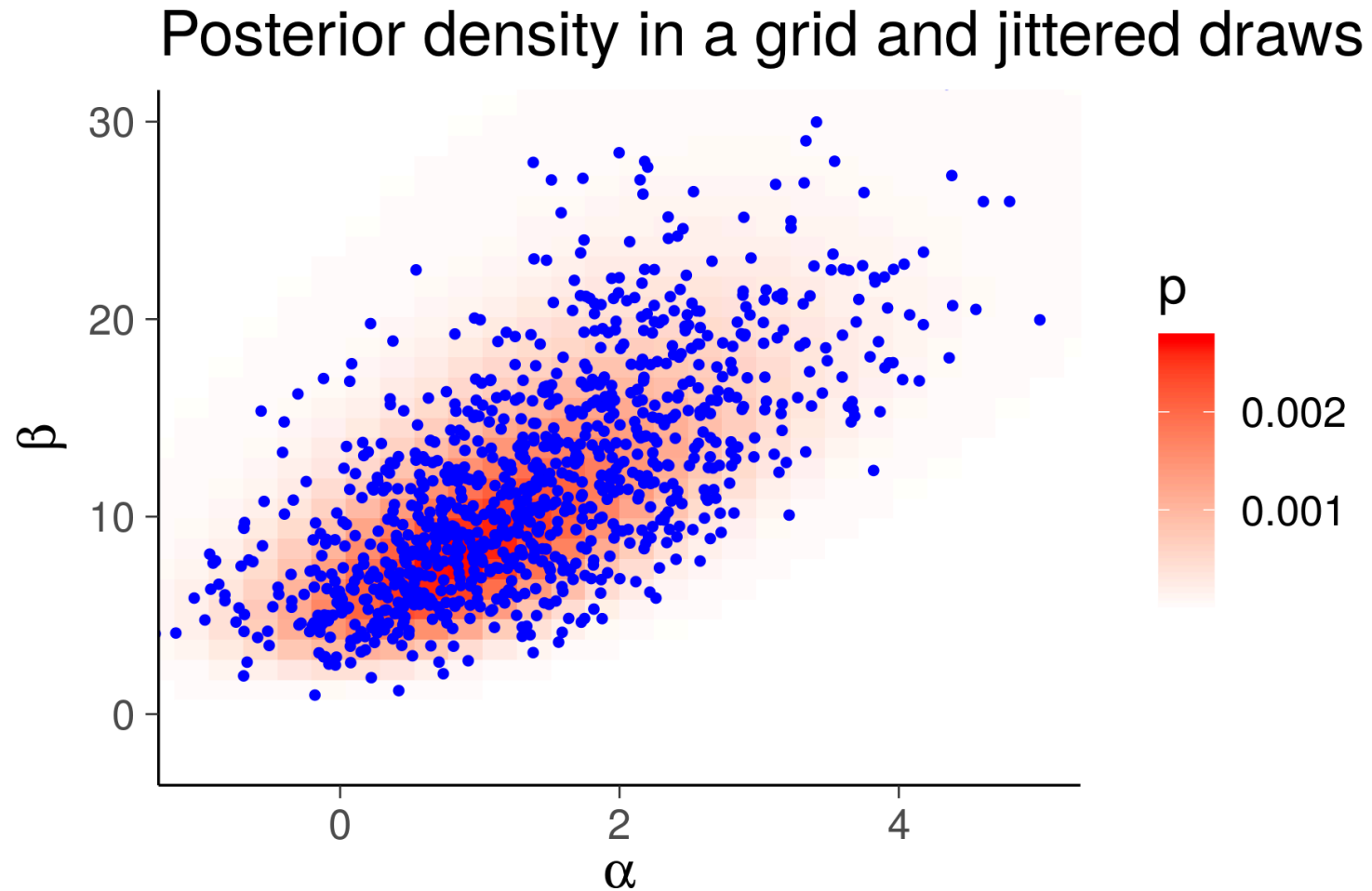
# Bioassay - grid

- Several draws can be from the same grid cell



# Bioassay - grid

- Jitter can be added to improve visualization



# Grid sampling

- Draws can be used to estimate expectations, for example

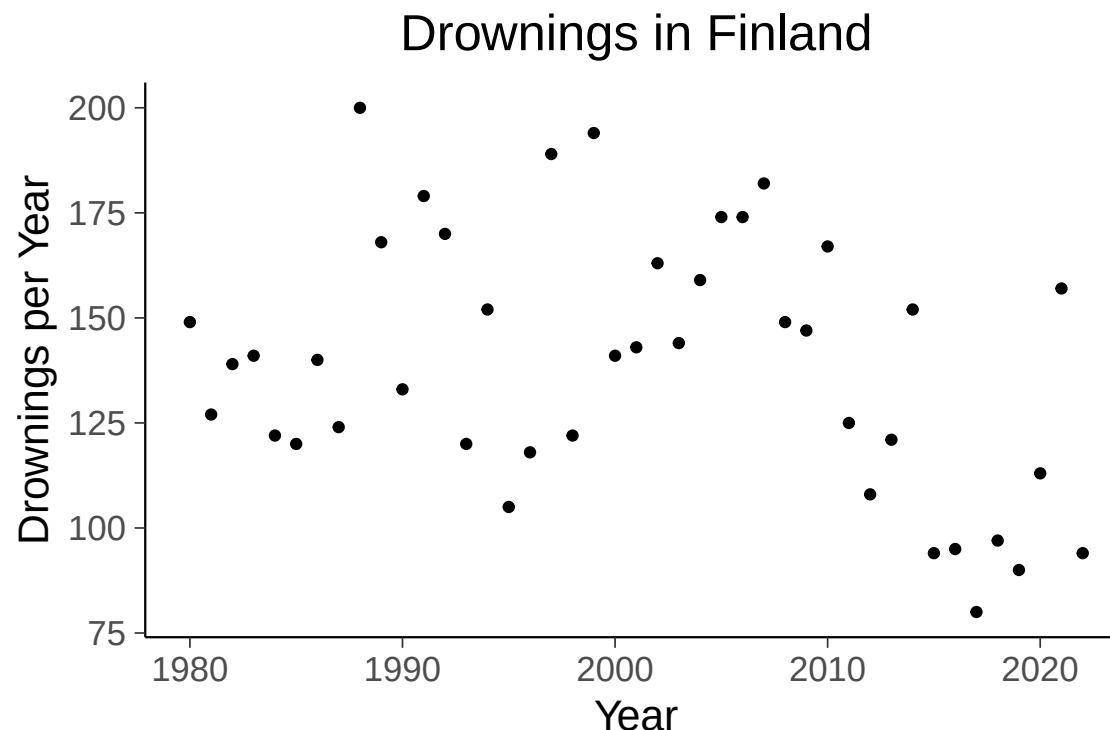
$$E[x_{\text{LD50}}] = E\left[-\frac{\alpha}{\beta}\right] \approx \frac{1}{S} \sum_{s=1}^S -\frac{\alpha^s}{\beta^s}$$

- Instead of sampling, grid could be used to evaluate functions directly, for example

$$E\left[-\frac{\alpha}{\beta}\right] \approx \sum_{t=1}^T -\frac{\alpha^t}{\beta^t} w_{\text{cell}}^t$$

where  $w_{\text{cell}}^t$  is the normalized probability of a grid cell  $t$ , and  $\alpha^t$  and  $\beta^t$  are center locations of grid cells

# Example: GLM for counts



- On average ~140 drownings per year
- Finnish government has invested in measures for reducing deaths
- Recent narrative based on effectiveness of education

Bayesian methods help

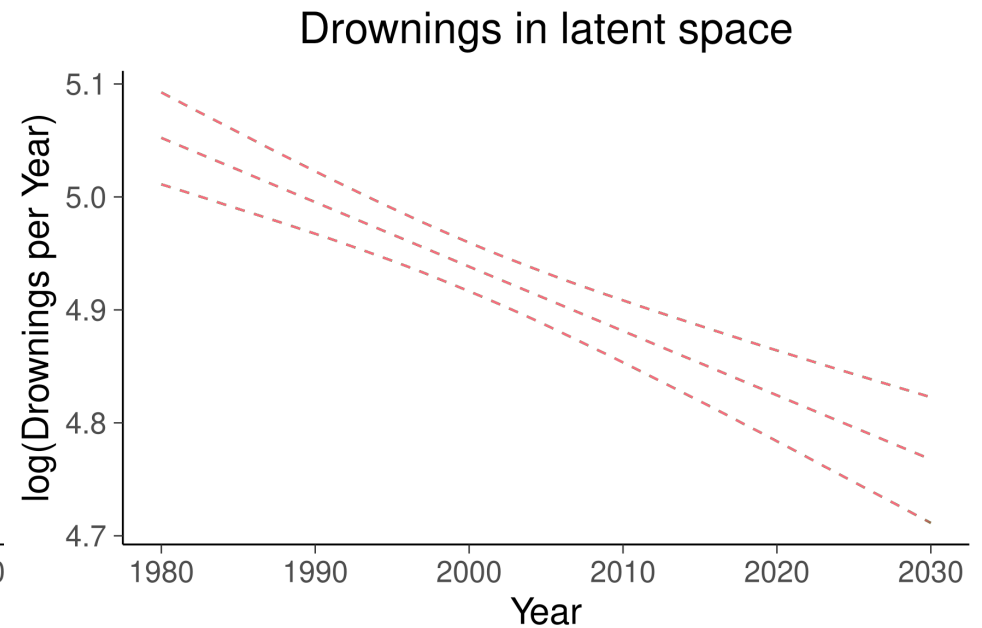
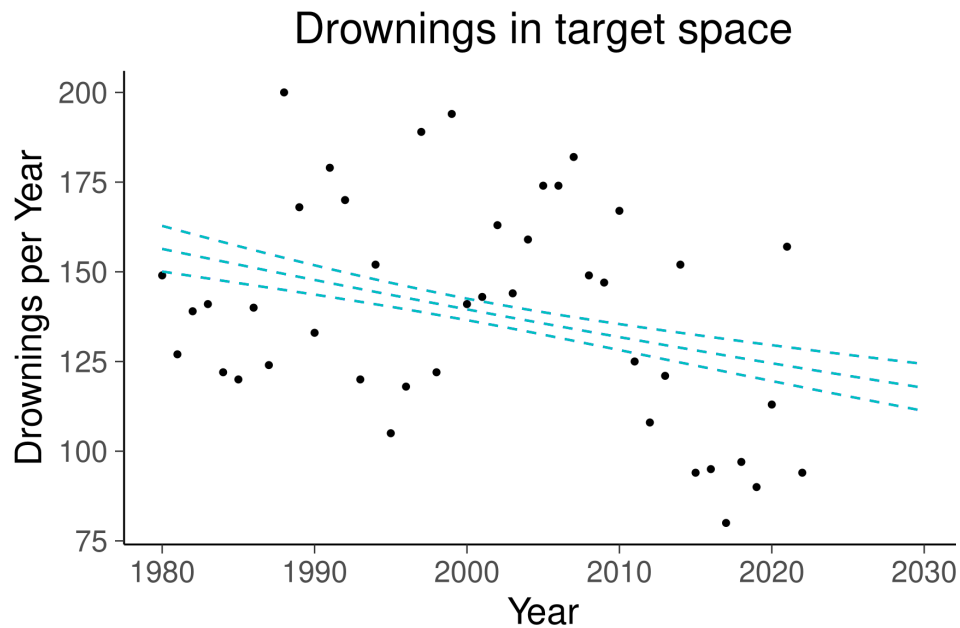
- Describe trends over time
- Evaluate uncertainty

# Example: GLM for counts

- Poisson regression model for counts:

$$y_i \mid \mu_i \sim \text{Poisson}(\mu_i)$$

$$\mu_i = e^{\alpha + \beta x_i}$$

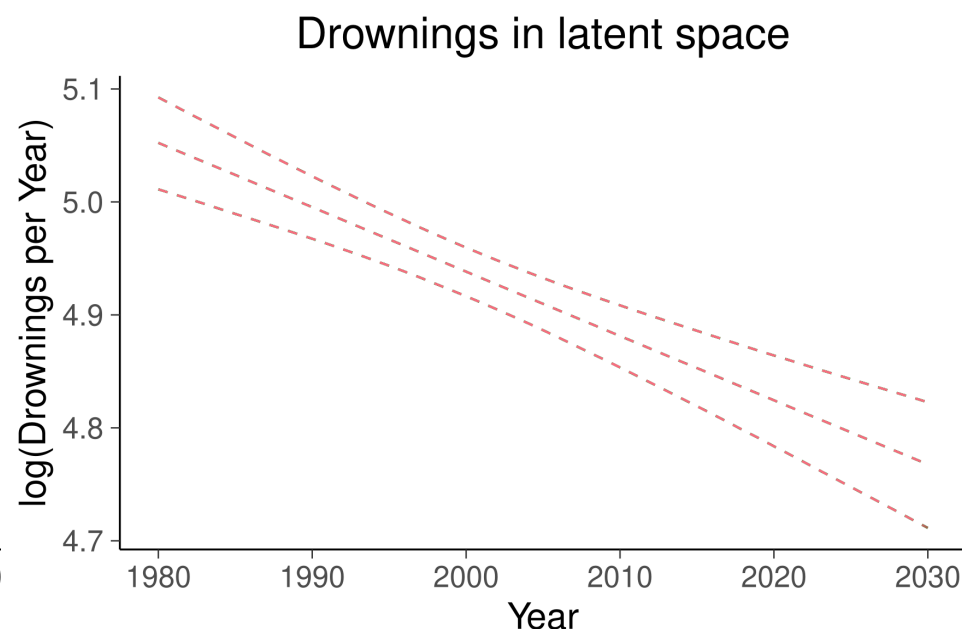
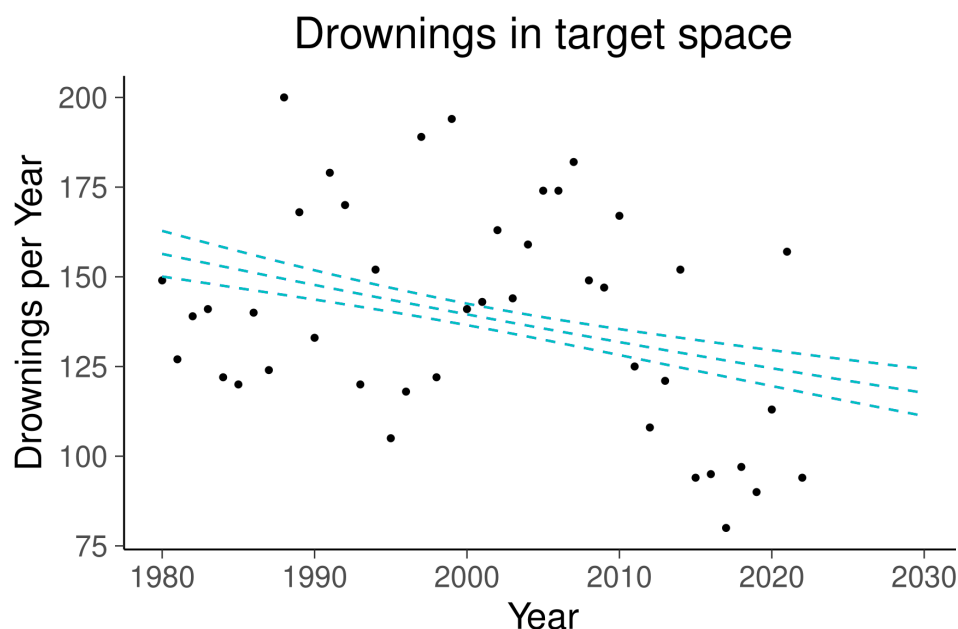


# Example: GLM for counts

- Poisson regression model for counts:

$$y_i \mid \mu_i \sim \text{Poisson}(\mu_i)$$

$$\mu_i = e^{\alpha + \beta x_i}$$



- Alternatively, Negative Binomial regression can be used for overdispersed count data.

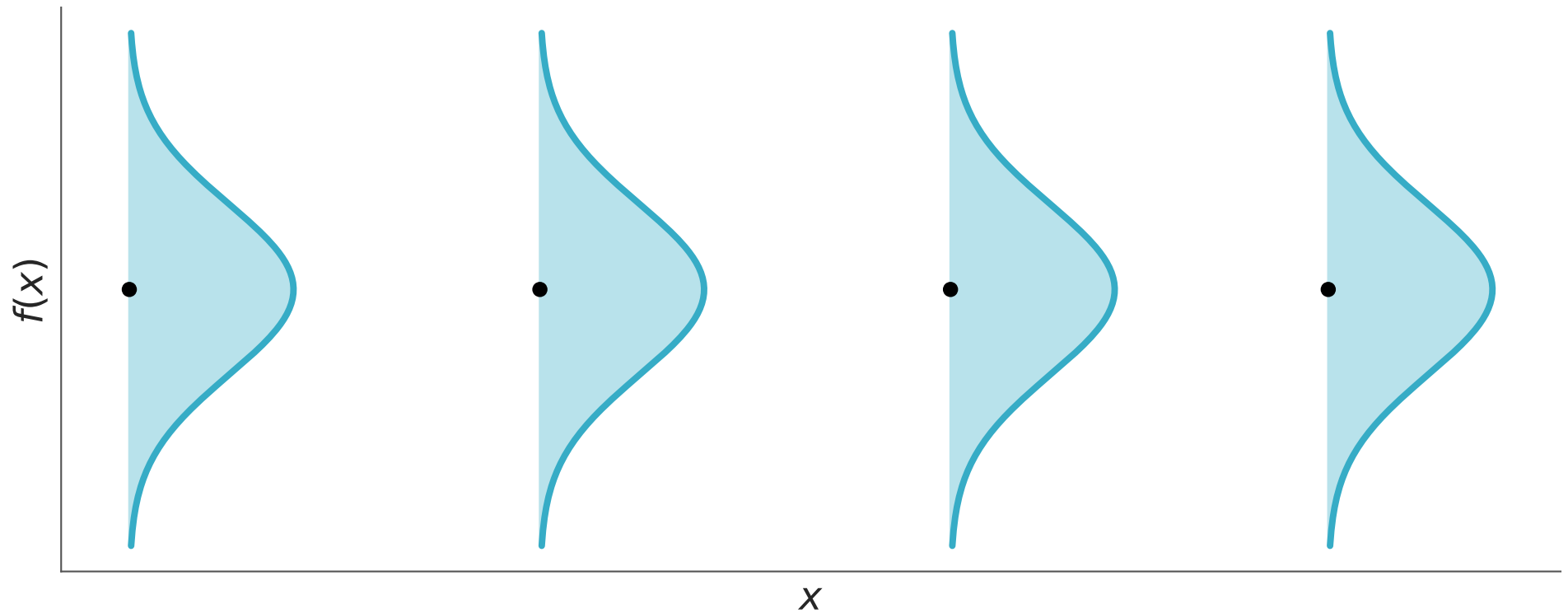
$$y_i \mid \mu_i, \varphi \sim \text{Neg-bin}(y_i \mid \mu_i, \varphi)$$

# Gaussian processes

- Imagine we want to represent a function in probabilistic terms

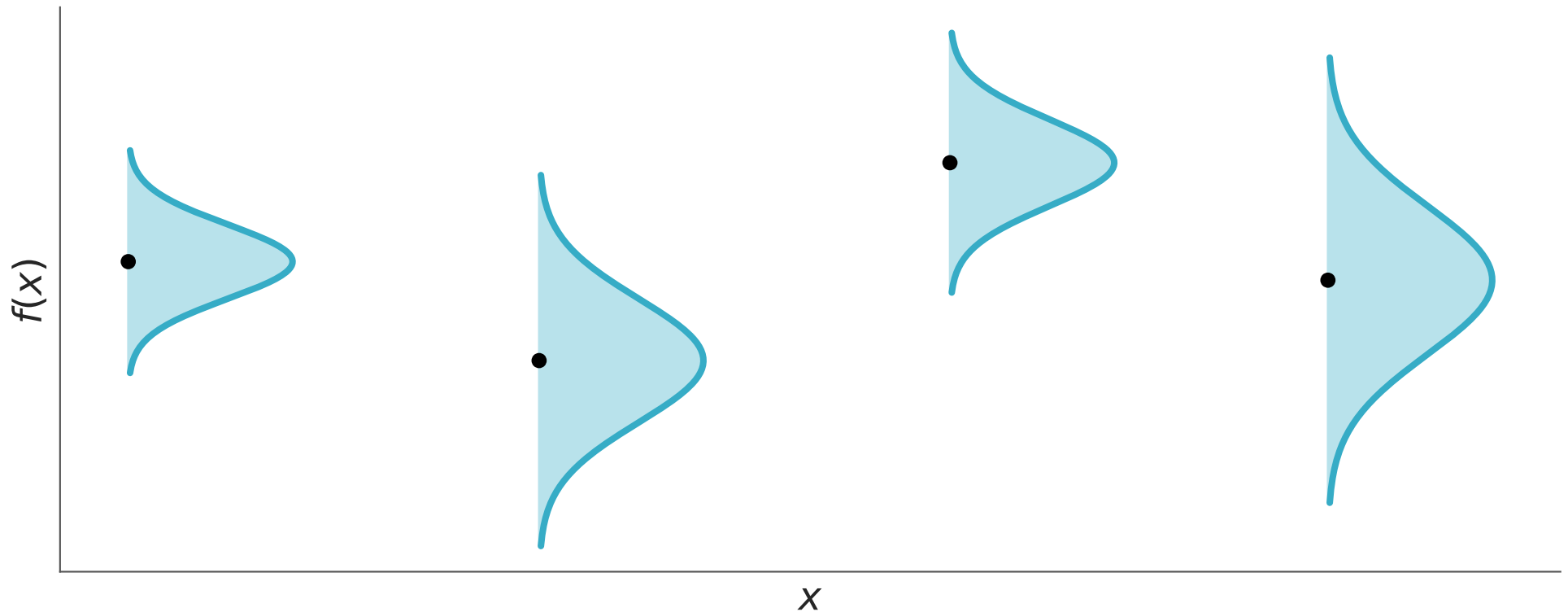
# Gaussian processes

- Imagine we want to represent a function in probabilistic terms
- We could associate a Gaussian with each function value.



# Gaussian processes

- We can vary the mean and variance of the Gaussian associated with each function value.

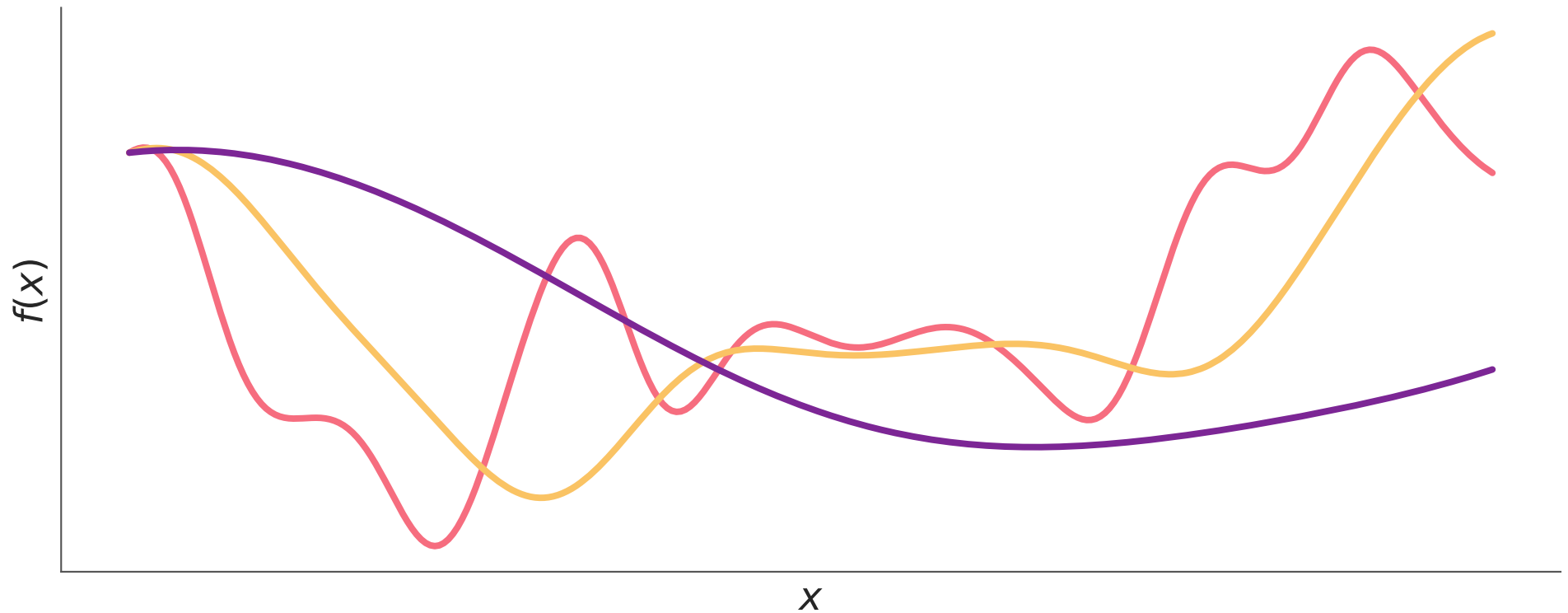


# Gaussian processes

- Instead of independent normals, we use a Multivariate normal distribution to model the joint behavior of all function values.

# Gaussian processes

- Instead of independent normals, we use a Multivariate normal distribution to model the joint behavior of all function values.
- This way, we introduce correlations between function values at different points.



# Gaussian processes

- Formally, it's an infinite-dimensional multivariate normal - any finite set of function values,  $f(x_1), \dots, f(x_n)$ , is jointly normal.

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# Gaussian processes

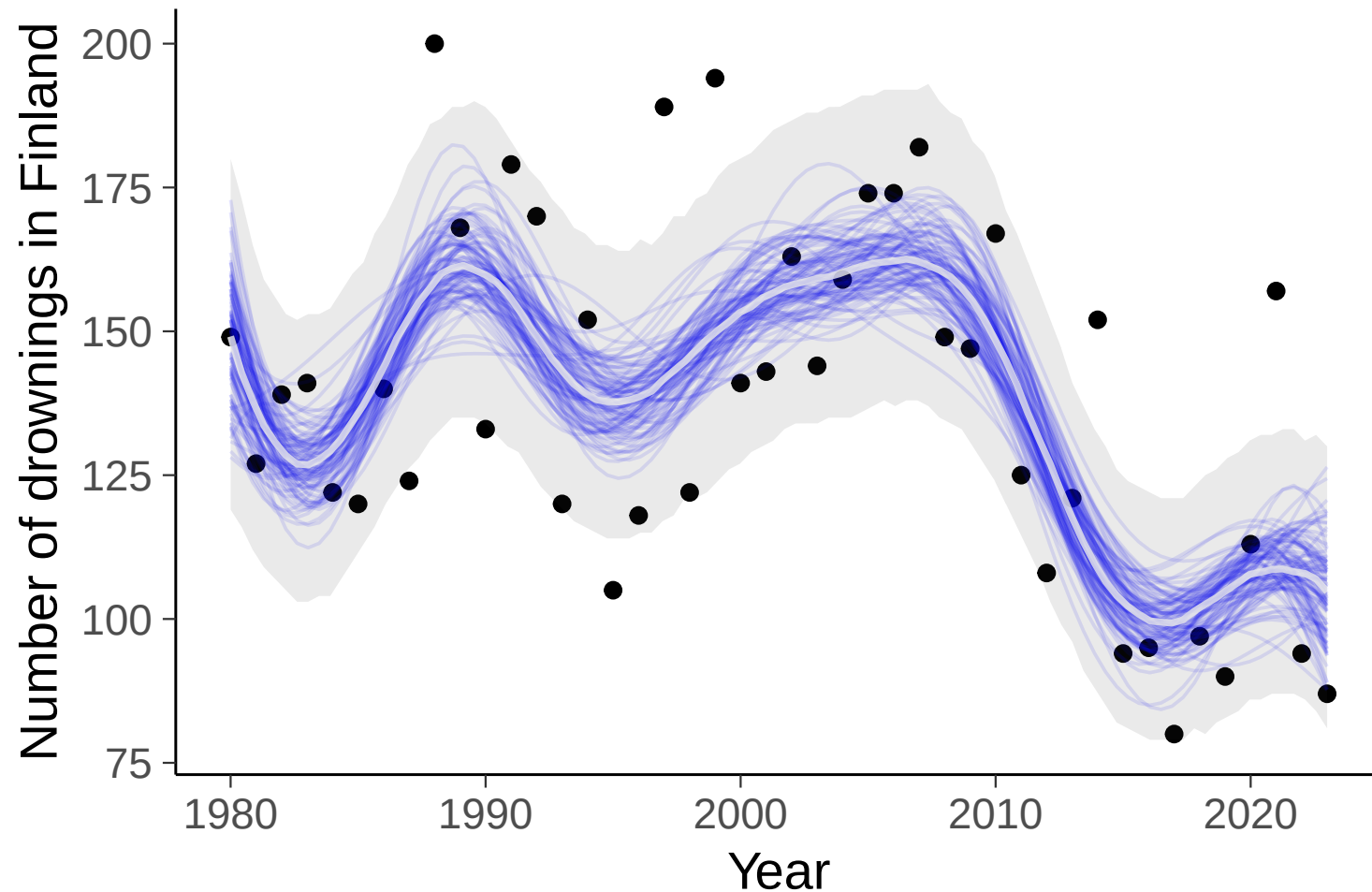
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- In practice we only ever evaluate a function at **finitely** many points → Multivariate normal
- Some nice theoretical properties and many models can be seen as special cases of Gaussian processes.
- More about GPs
  - See [Gaussian Processes for Machine Learning](#)
  - GPs in BDA3 Chapter 21
  - Other courses in Aalto.

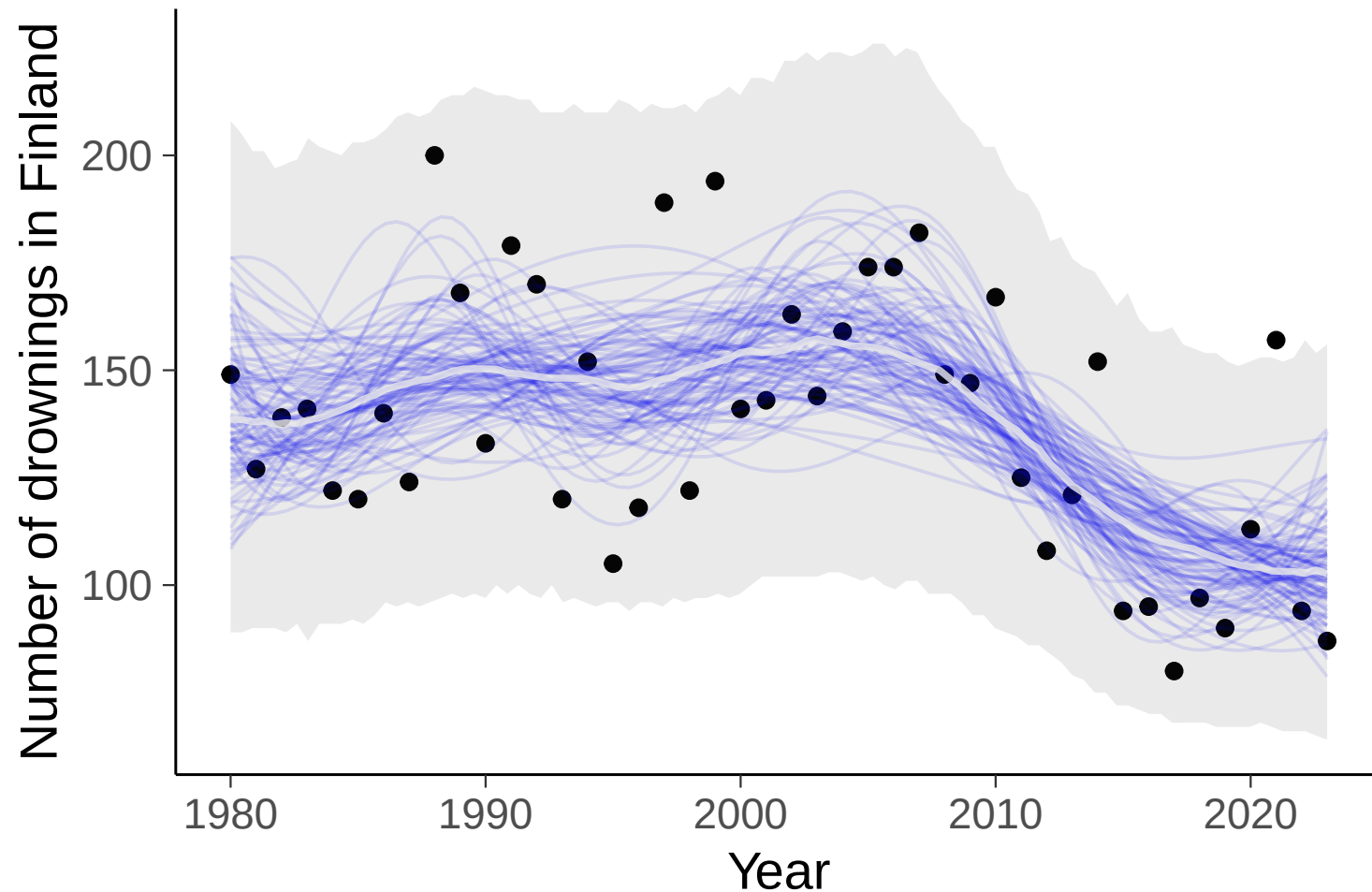
# Example GLM: Gaussian Process Models

$$y_i \mid \mu_i \sim \text{Poisson}(\mu_i) \quad \mu_i \sim e^{f_i}, \quad f \sim \text{GP}(0, k(\text{Year}, \theta))$$



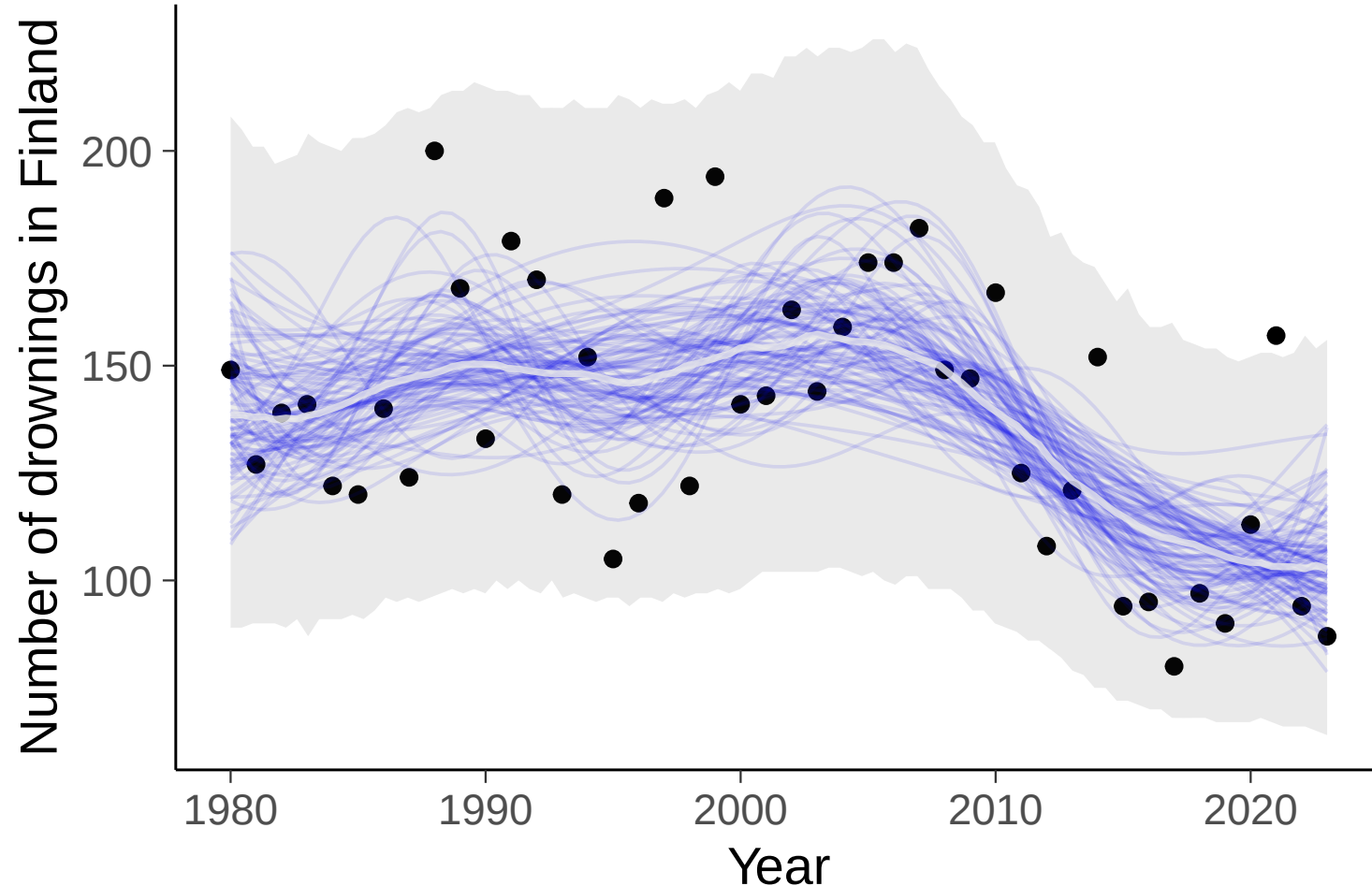
# Example GLM: Gaussian Process Models

$$y_i \mid \mu_i \sim \text{Neg-bin}(\mu_i, \varphi) \quad \mu_i \sim e^{f_i}, \quad f \sim \text{GP}(0, k(\text{Year}, \theta))$$



# Example GLM: Gaussian Process Models

- NegativeBinomial model is better because data is overdispersed
  - later in the course we will learn about tools to assess model fit and compare models
- Trend interpretations shouldn't be based on one observation



# Thinking counts

- For simplicity of exposition, we often start learning with normal observation models
- But we observe count data on a daily basis
- Very relevant in industry (number of sold products, ad views, customer count, etc.)
- Can you think of such examples from the classroom?
  - Think of how many students attend BDA lectures over the course
  - Number of students who report getting sick over time until Christmas
  - Number of dropouts
  - Would you expect overdispersion?



# Appendix A1: derivation of the sum identity

We now show the identity used above,  $\sum_i (y_i - \mu)^2 = \sum_i (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$ , by expanding the square:

$$\begin{aligned} \sum_{i=1}^n (y_i - \mu)^2 \\ = \sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2) \end{aligned}$$

We add and subtract  $\bar{y}^2 - 2y_i\bar{y}$  inside the sum — this doesn't change the value, but lets us regroup terms around  $\bar{y}$  instead of  $\mu$ :

$$= \sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2 - \bar{y}^2 + \bar{y}^2 - 2y_i\bar{y} + 2y_i\bar{y})$$

# Appendix A1: derivation of the sum identity (cont)

Grouping the terms that involve only  $y_i$  and  $\bar{y}$  into one sum, and the remaining terms into another:

$$= \sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) + \sum_{i=1}^n (\mu^2 - 2y_i\mu - \bar{y}^2 + 2y_i\bar{y})$$

The first sum is exactly  $\sum_i (y_i - \bar{y})^2$ . In the second sum, evaluating the summation using  $\sum_{i=1}^n y_i = n\bar{y}$  gives  $\sum_{i=1}^n (-2y_i\mu + 2y_i\bar{y}) = -2n\bar{y}\mu + 2n\bar{y}^2$ . Factoring out  $n$ :

$$= \sum_{i=1}^n (y_i - \bar{y})^2 + n(\mu^2 - 2\bar{y}\mu - \bar{y}^2 + 2\bar{y}^2)$$

The bracket is a perfect square,  $(\bar{y} - \mu)^2$ :

$$= \sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$$

# Appendix A2: Marginal posterior $p(\mu \mid y)$ derivation

We now derive  $p(\mu \mid y)$  analytically, by integrating the joint posterior over  $\sigma^2$ :

$$\begin{aligned} p(\mu \mid y) &= \int_0^\infty p(\mu, \sigma^2 \mid y) d\sigma^2 \\ &\propto \int_0^\infty \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) d\sigma^2 \end{aligned}$$

To evaluate this, substitute a new variable  $z$  that absorbs the bracket term and turns the integral into a recognizable form:

# Appendix A2: Marginal posterior $p(\mu \mid y)$ derivation (cont)

Transformation

$$A = (n - 1)s^2 + n(\mu - \bar{y})^2 \quad \text{and} \quad z = \frac{A}{2\sigma^2}$$

$$p(\mu \mid y) \propto A^{-\frac{n}{2}} \int_0^\infty z^{\frac{n-2}{2}} \exp(-z) dz$$

Recognize gamma integral  $\Gamma(u) = \int_0^\infty x^{u-1} \exp(-x) dx$

The integral is now just a constant (a gamma function value not depending on  $\mu$ ), leaving only  $A$ 's dependence on  $\mu$ :

$$\propto [(n - 1)s^2 + n(\mu - \bar{y})^2]^{-\frac{n}{2}}$$

# Appendix A2: Marginal posterior $p(\mu \mid y)$ (cont)

Dividing through by  $(n - 1)s^2$  to isolate the standard Student- $t$  kernel form:

$$\propto \left[ 1 + \frac{n(\mu - \bar{y})^2}{(n - 1)s^2} \right]^{-\frac{n}{2}}$$

$$p(\mu \mid y) = t_{n-1} \left( \mu \mid \bar{y}, \frac{s^2}{n} \right) \text{ Student's } t$$